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BESIII Analysis Memo

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BESIII Analysis Memo (May 15, 2026)

Cross section measurement of process $e^+e^- \to \pi^+\pi^-\pi^0$ from threshold to 2.0 GeV via ISR technique

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Abstract

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List of changes

For an easy reading of the document, major changes in various versions with respect to the previous one are listed below.

Version 1.0

- First version of memo.

Contents

1 Introduction **3**

2 BEPCII and BESIII detector **4**

3 Data and MC samples **5**

4 Event Selections **7**

5 Background Study **9**
5.1 Estimate of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ . . . 14
5.2 Estimate of Other Backgrounds and Signal Yields . . . 17

6 Correction of Signal Efficiency **21**
6.1 Tracking and PID efficiencies . . . 22
6.2 The π^0 reconstruction efficiency . . . 22
6.3 The ISR photon detection efficiency . . . 25
6.4 PHOKHARA Extra Photon Fraction . . . 29
6.4.1 Large Angle NLO Photons . . . 30
6.4.2 Small Angle NLO Photons . . . 31
6.4.3 Data-MC Comparison of NLO Photon Fractions . . . 32
6.5 Correcting the signal yields . . . 34

7 Unfolding Procedure **36**
7.1 The IDS unfolding method . . . 36
7.2 Parameter optimization and Unfolding . . . 37

8 Systematic Uncertainties **42**
8.1 Efficiency correction uncertainties . . . 42
8.2 The detector resolution effects in unfolding . . . 44
8.3 The requirement on E/p . . . 45
8.4 The 4C kinematic fit . . . 46
8.5 Background estimate . . . 47
8.6 The π^0 veto . . . 47

9 Summary **49**
9.1 Calculation of cross sections . . . 49
9.2 Combination of independent dressed cross section measurements . . . 51
9.2.1 Formalism of BLUE . . . 51
9.2.2 Optimal weights . . . 51
9.2.3 Application to this analysis . . . 52

1 Introduction

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2 BEPCII and BESIII detector

The Beijing Electron-Positron Collider II (BEPCII) —a double-ring multi-bunch collider —is operating in the τ-charm energy region (2 to 4.9 GeV) with the design luminosity of 1×10^{33} cm^{-2}s^{-1} [2]. The Beijing Spectrometer III (BESIII) operating at BEPCII is a high-precision, general-purpose detector that records symmetric e^+e^- collisions provided by the BEPCII storage ring. BESIII is approximately cylindrically symmetric with 93% coverage of the solid angle around the e^+e^- interaction point (IP). The components of the apparatus ordered by distance from the IP, are a 43-layer small-cell main drift chamber (MDC), a time-of-flight (TOF) system based on plastic scintillators with two layers in the barrel region and one layer in the end-cap region (the end-cap TOF system has been updated with the multi-gap resistive plate chamber in 2014 [1]), a 6240-cell CsI(Tl) crystal electromagnetic calorimeter (EMC), a superconducting solenoid magnet providing a 1.0 T magnetic field aligned with the beam axis, and resistive plate muon-counter layers interleaved with steel. The momentum resolution for charged tracks in the MDC is 0.5% for transverse momenta with 1 GeV/c. The energy resolution in the EMC is 2.5% in the barrel region and 5.0% in the end-cap region for 1 GeV photons. For charged tracks, particle identification (PID) for charged tracks combines measurements of the energy deposit dE/dx in MDC and flight time in TOF and forms likelihoods $\mathcal{L}(h)(h = K, \pi)$ for a hadron h hypothesis. More details about the BESIII detector are provided in Ref [3].

Table 1: Beam energies and integrated luminosities of the data samples used in this work, quoted from Ref. [4, 5].

E_{beam}(GeV)	Integrated luminosity (pb^{-1})	Data taking time	RunNo
3.773	$2931.8 \pm 0.2 \pm 13.8$ [4]	2010(round03) 2011(round04)	11414 – 13988, 14395 – 14604 20448 – 23454
3.773	4995.0 ± 19.0 [5]	2022(round15)	70522 – 73929
3.773	8157.0 ± 31.0 [5]	2023(round16)	74031 – 78536
3.773	4191.0 ± 16.0 [5]	2024(round17)	78615 – 81094

Table 2: Processes and luminosity scales of the inclusive MC samples used in this analysis. Separate samples are available for each data-taking round. Most processes use samples with an integrated luminosity equivalent to ten times that of the data.

Process	Luminosity scale
$e^+e^- \to \psi(3770) \to D^0\bar{D}^0$	$10\times$
$e^+e^- \to \psi(3770) \to D^+D^-$	$10\times$
$e^+e^- \to \tau^+\tau^-$	$10\times$
$e^+e^- \to \psi(3770) \to$ non-$D\bar{D}$	$10\times$
$e^+e^- \to q\bar{q}$	$10\times$
$e^+e^- \to \gamma_{\mathrm{ISR}}\psi(2S)$	$10\times$
$e^+e^- \to \gamma_{\mathrm{ISR}}J/\psi$	$10\times$
$e^+e^- \to e^+e^-$	$0.25\times$
$e^+e^- \to \gamma\gamma$	$3\times$
$e^+e^- \to \mu^+\mu^-$	$15\times$

3 Data and MC samples

We use an e^+e^- collision sample corresponding to an integrated luminosity of 20 fb^{-1}, collected by BESIII detector at center-of-mass energy of 3.773 GeV, under BOSS-7.1.2 version. The details are listed in Table 1, quoted from Ref. [4, 5].

Simulated Monte Carlo (MC) samples are employed to study backgrounds, optimize event selection, and estimate detection efficiencies. Two categories of MC samples are utilized in this analysis:

- **The inclusive MC samples** comprehensively model known physics processes at $\sqrt{s} = 3.773$ GeV, including $D\bar{D}$ production, $\tau^+\tau^-$ pair production, initial-state-radiation (ISR) production of vector charmonium states (J/ψ and $\psi(2S)$), QED processes, and continuum light hadron production. The charm working group provides separate samples for the different data-taking rounds(round03-04, round15, round16, and round17) [7]. In this analysis, we use inclusive MC samples with an integrated luminosity equivalent to ten times that of the data for most processes, except for the QED processes $e^+e^- \to e^+e^-$, $\gamma\gamma$, and $\mu^+\mu^-$, where we use the samples with the scaling factors provided by the charm working group. The processes and the corresponding luminosity scales of the samples used in this analysis are summarized in Table 2.

- **Exclusive signal and background MC samples** generated specifically for this analysis using the PHOKHARA event generator [8, 9, 10] (version 9.1). These samples are produced independently for each data-taking round (round03, round04, round15, round16, round17) to account for variations in detector conditions. Sufficient statistics are generated for each process to ensure reliable efficiency determination and background studies. The generated processes include:
  - $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}$ (signal process)
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$
  - $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\mathrm{ISR}}$

  These exclusive samples are used for signal efficiency studies, background characterization, and validation of the analysis methodology.

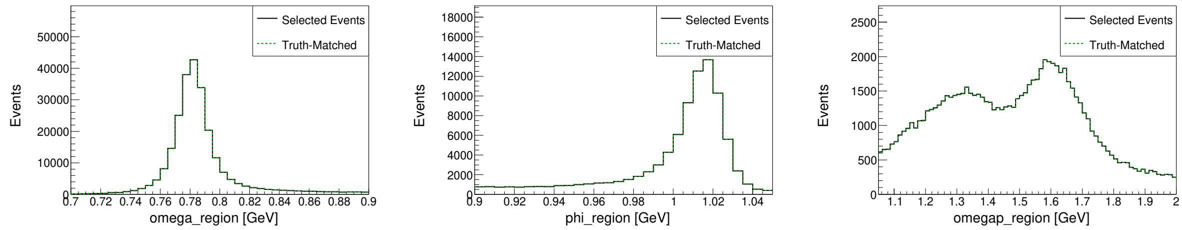
4 Event Selections

The process under investigation is $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$, whose final states include two charged tracks, one neutral π^0, and one ISR photon. The following selection criteria is applied to all the data samples. Here the figures are shown by taking $\sqrt{s}$ = 3.773GeV round16 data as an example. The distributions related to the event selection of other data sets are shown in Appendix. XXX. Situations are similar to each other.

- Good charged track: A charged track is taken as a good MDC track if it originates from the beam vertex and lies within the MDC polar angle region:
  - $|\cos\theta| < 0.93$, where θ is the polar angle with respect to the beam axis.
  - $|\Delta r| < 1$ cm, $|\Delta z| < 10$ cm, where $|\Delta r|$ and $|\Delta z|$ is distance of closest approach to the interaction point (IP) perpendicular to the beam axis and along the beam axis, respectively.
  - the ratio of the calorimeter-deposited energy to the track momentum E_{EMC}/p is required to be less than 0.8 for both of the charged tracks.
  - no PID requirements for π^+, π^-.
- γ selection:
  - $E > 0.025$ GeV (barrel: $|\cos\theta| < 0.8$), $E > 0.050$ GeV (endcap: $0.86 < |\cos\theta| < 0.92$).
  - $0 \leqslant t_{\rm TDC} \leqslant 14(\times 50\text{ns})$ to suppress fake photons due to electronic noise or beam background.
  - the closest angle with all charged tracks $\theta > 10^\circ$ to suppress background from radiation of charged tracks.
- $\pi^0 \to \gamma\gamma$ selection:
  - two photon invariant mass $M_{\gamma\gamma}$ within (0.115, 0.150) GeV$/c^2$.
  - To improve the momentum resolution, 1C kinematic fit is performed. Here, the 1C means π^0 mass constraint, and the updated 4-momentum will be used in the further analysis. $\chi^2_{\rm KF} < 10$

Candidate events are selected by requiring exactly two good charged tracks, accompanied by at least one reconstructed π^0 and a minimum of three good photon showers. This ensures the presence of at least one π^0 and one isolated photon from Initial-State Radiation ($\gamma_{\rm ISR}$). A four-constraint (4C) kinematic fit is then applied to the event, and the candidate with the smallest $\chi^2_{\rm 4C}$ value is chosen for further analysis.

After identifying an ISR photon from the isolated photon showers, the invariant mass of the remaining $e^+e^- \to \pi^+\pi^-\pi^0$ system is reconstructed. As expected for ISR processes, the resulting center-of-mass energy $\sqrt{s'}$ of the $\pi^+\pi^-\pi^0$ system shows a continuous spectrum, characteristic of the radiative return mechanism. This continuum behavior is illustrated in Fig. 1 across the $\omega(782)$, $\phi(1020)$, and ω'/ω'' mass regions.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω' and ω'' region

Figure 1: Invariant mass spectra of MC in the (a) $\omega(782)$, (b) $\phi(1020)$, and (c) ω'/ω'' mass regions. The distributions correspond to $M(pK_S^0)$, $M(p\pi^0)$, and $M(K_S^0\pi^0)$, respectively. The "Truth-Matched" distribution (green dashed line) corresponds to reconstructed events where the momentum directions of the π^0 and $\gamma_{\rm ISR}$ successfully match their truth-level counterparts. A loose requirement of $\chi^2_{\rm KF} < 100$ is applied to the kinematic fit. The continuous spectrum of $\sqrt{s'}$ for the $\pi^+\pi^-\pi^0$ system demonstrates the radiative return mechanism via ISR.

5 Background Study

In the study of background processes, the main backgrounds are categorized into $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$, QED processes, and others. The dominant contributions arise from processes with additional π^0, particularly $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, while $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$ is not dominant but shares the same underlying mechanism as a background to our signal. In the inclusive analysis, the cross sections used for these two processes were not correctly configured.

To verify that these processes constitute the primary backgrounds, the $\mathcal{L} \times \sigma_{\rm hadron}$ method is employed, where $\mathcal{L}$ is the integrated luminosity and $\sigma_{\rm hadron}$ is the hadronic cross section. The luminosity values for each data-taking period are listed in Table 1. The background yield is estimated as a function of the $\pi^+\pi^-\pi^0$ invariant mass (m) using the effective luminosity spectrum, which is related to the mass spectrum via:

$$\frac{dL_{\rm eff}}{dm} = \frac{2m}{s} \cdot F(s,m) \cdot \mathcal{L}_{\rm int}, \tag{1}$$

where $s = 4E^2$, E is the beam energy, $\mathcal{L}_{\rm int}$ is the total integrated luminosity, and $F(s,m)$ is the probability function for initial-state photon emission. This function can be written as [11]:

$$\beta = \frac{2\alpha}{\pi}\left[\ln\left(\frac{s}{m_e^2}\right) - 1\right], \tag{2}$$

$$x = 1 - \frac{m^2}{s}, \tag{3}$$

$$\begin{aligned} \frac{dL_{\rm eff}}{dm} = \frac{2m}{s} \cdot \Bigg\{ &\beta x^{\beta-1}\left[1 + \frac{\alpha}{\pi}\left(\frac{\pi^2}{3} - \frac{1}{2}\right) + \frac{3}{4}\beta \right. \\ &\left. + \beta^2\left(\frac{37}{96} - \frac{\pi^2}{12} - \frac{1}{72}\ln\frac{s}{m_e^2}\right)\right] \\ &- \beta\left(1 - \frac{x}{2}\right)\ln\frac{1}{x} - \frac{1+3(1-x)^2}{x}\ln(1-x) - 6 + x\Bigg\} \cdot \mathcal{L}_{\rm int}, \end{aligned} \tag{4}$$

where $\alpha = 1/137$ is the fine-structure constant, $m_e = 0.5110034 \times 10^{-3}$ GeV$/c^2$ is the electron mass, and $\sqrt{s} = 3.773$ GeV is the center-of-mass energy of the primary e^+e^- system.

The effective luminosity spectrum calculated using this formula for round16 is shown in Fig. 2. Similar calculations were performed for all data-taking rounds.

Based on this method, rough yield estimates for the two processes with additional π^0, $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ (non-ISR) and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$ (ISR), were calculated for the four data-taking periods (round03/04, round15, round16, round17). The results are summarized in Table 3. For the ISR channel, the estimation relies on the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section measured by the BaBar collaboration [12].

It is important to note that the values listed in the table are initial yield estimates without any selection efficiency corrections; the actual background contribution to the signal region is obtained by multiplying these yields by the corresponding selection efficiency relevant to the specific analysis cuts. These estimates serve only to confirm that these processes constitute the main sources of background and to provide guidance for optimizing subsequent selection criteria.

The final background yield is then obtained by multiplying the effective luminosity spectrum by

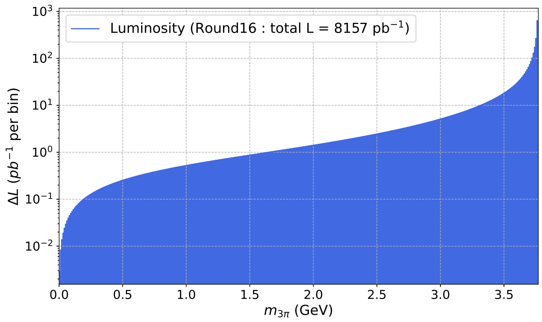


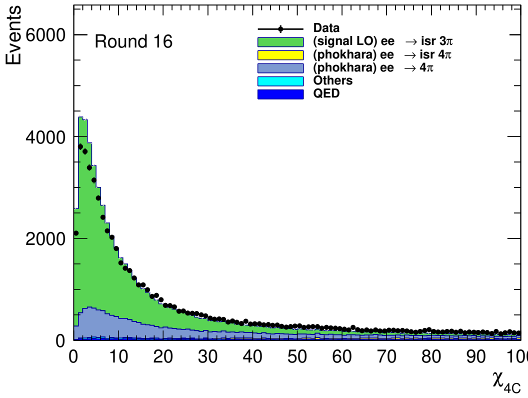
Figure 2: Effective luminosity spectrum as a function of the $\pi^+\pi^-\pi^0$ invariant mass ($m_{3\pi}$) calculated for round16 with total integrated luminosity $\mathcal{L}_{\rm int} = 8157$ pb^{-1}. The spectrum is shown with logarithmic scale on the vertical axis.

Table 3: Rough estimates of the expected yields for the two processes with additional π^0 before applying any selection efficiency. The integrated luminosities are taken from Table 1. The cross section for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ is taken as $\sigma = 0.22$ nb at $\sqrt{s} = 3.773$ GeV, while the energy-dependent cross section for the ISR process is derived from the measured $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ cross section [12]. The final background contribution is obtained by multiplying these yields by the corresponding selection efficiency.

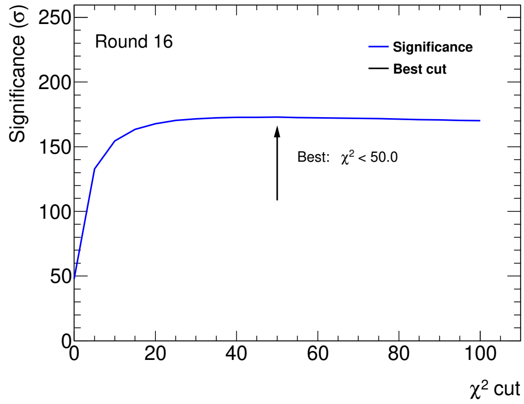
Data sample (Run period)	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0)$	$N(e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR})$
round03-04	511940	1142096
round15	872209	1945825
round16	1424346	3177597
round17	731817	1632623

the cross section and a selection-efficiency factor that varies dynamically with the analysis cuts. This method is used solely to confirm that $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ and $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$ are the dominant backgrounds and to optimize the selection criteria. It is not used to derive the final background estimates; precise yields for these two components will be determined using data-driven methods. Based on the rough background estimate from $\mathcal{L} \times \sigma_{\rm hadron}$, the signal yield was approximated by subtracting the background from the data, allowing for a preliminary significance calculation. Using this approach, a figure of merit (FOM) defined as $S/\sqrt{S+B}$ was constructed to evaluate the signal significance. To optimize the event selection, a scan over the kinematic fit χ^2 was performed, with the result for Round 16 shown in Fig. 3. The Round 16 optimization yields an optimal cut of $\chi^2_{\rm KF} < 50$. As shown in Fig. 4, consistent convergence of significance is also observed for Rounds 03/04, 15, and 17 with the same cut.

Based on the optimal $\chi^2_{\rm KF}$ selection, the invariant mass spectrum of the 3π system ($M_{3\pi}$) can be divided into three distinct regions for further study: the $\omega(782)$ region, the $\phi(1020)$ region, and a higher-mass region associated with possible ω'/ω'' excitations. In the high-mass ω'/ω'' region, a significant background component from events containing additional π^0 mesons was observed. To quantify this, the number of π^0 candidates per event was counted, where any π^0 candidate satisfying a one-constraint kinematic fit $\chi^2 < 200$ was accepted. The distribution of this π^0 multiplicity is shown in Fig. 5. To

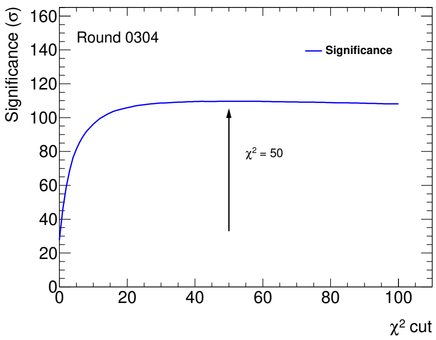


(a) χ^2 distribution for the 4C kinematic fit

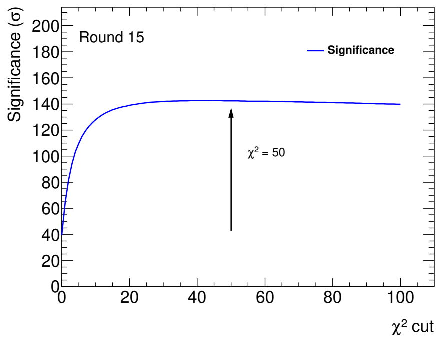


(b) FOM scan as a function of the χ^2 cut

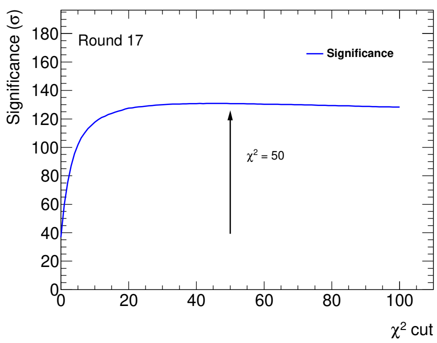
Figure 3: (a) Distribution of the kinematic fit χ^2 for the four-constraint (4C) fit using the Round 16 data sample. (b) The corresponding figure of merit $(S/\sqrt{S+B})$ as a function of the applied χ^2 cut value for the Round 16 data.



(a) Round 03/04



(b) Round 15



(c) Round 17

Figure 4: FOM $(S/\sqrt{S+B})$ as a function of the applied χ^2 cut value for (a) Round 03/04, (b) Round 15, and (c) Round 17 data samples.

suppress this specific background, a requirement of exactly one π^0 candidate per event was applied exclusively in the ω'/ω'' mass region. No such restriction was applied to the $\omega(782)$ and $\phi(1020)$ signal regions. The following studies and distributions are based on the Round 16 data sample.

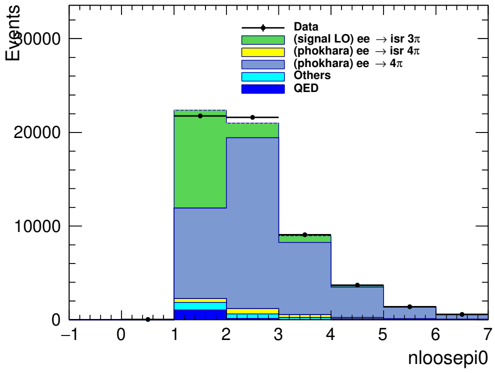
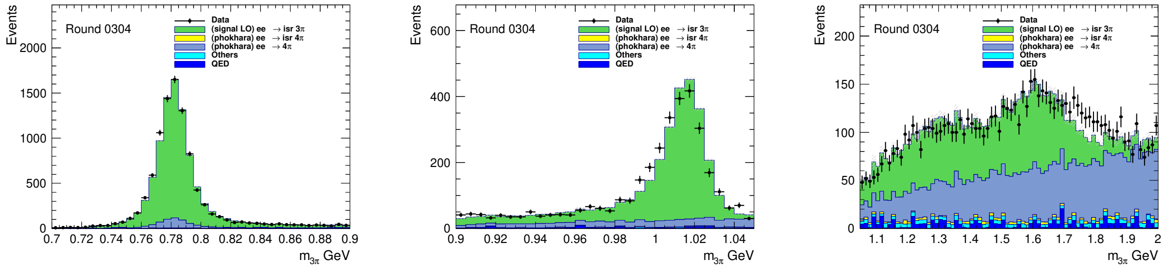
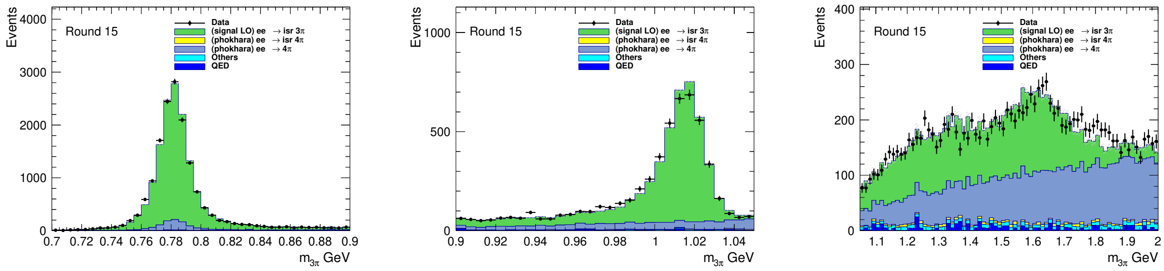


Figure 5: Distribution of the number of π^0 candidates per event in the ω'/ω'' mass region using the Round 16 data. A π^0 candidate is counted if its 1C kinematic fit $\chi^2 < 200$.

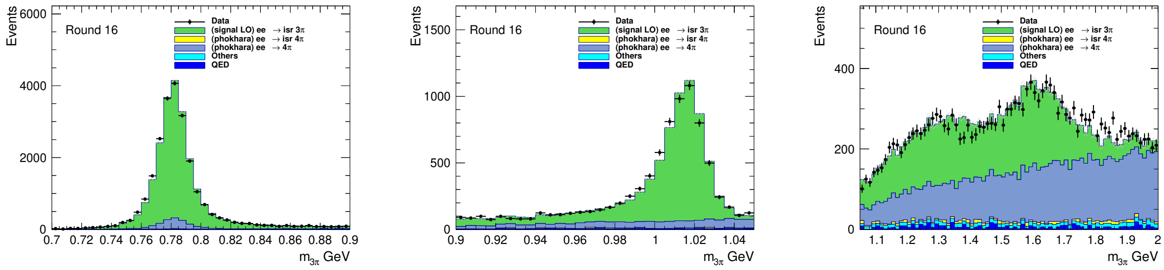
The resulting $M_{3\pi}$ invariant mass spectra, after applying all selection criteria including the region-specific π^0 multiplicity cut, are presented separately for the three mass regions in Fig. 6. The $e^+e^- \to \pi^+\pi^-\pi^0\pi^0\gamma_{\rm ISR}$ contribution is negligible and will not be treated as a separate background in the subsequent fitting procedure; only the non-ISR $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ process will be explicitly estimated using data-driven methods and constrained in the fit.



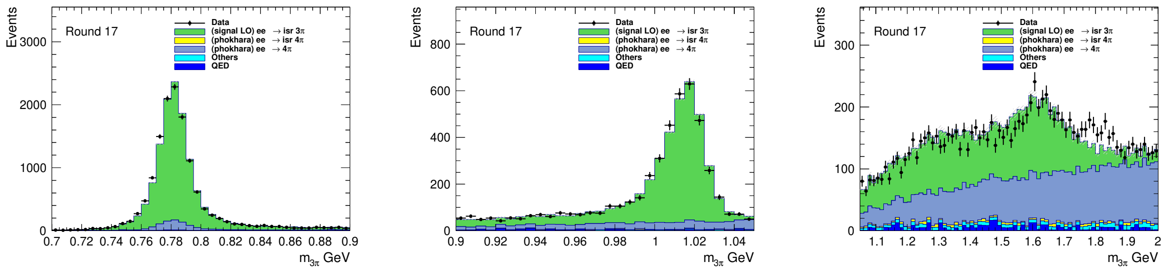
(a) Round 03/04, $\omega(782)$ (b) Round 03/04, $\phi(1020)$ (c) Round 03/04, ω'/ω''



(d) Round 15, $\omega(782)$ (e) Round 15, $\phi(1020)$ (f) Round 15, ω'/ω''



(g) Round 16, $\omega(782)$ (h) Round 16, $\phi(1020)$ (i) Round 16, ω'/ω''



(j) Round 17, $\omega(782)$ (k) Round 17, $\phi(1020)$ (l) Round 17, ω'/ω''

Figure 6: Invariant mass spectra $M_{3\pi}$ for different data rounds and mass regions with $N_{\pi^0} = 1$ (other columns no π^0 multiplicity cut).

5.1 Estimate of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$

In estimating the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ background, the following strategy was adopted:

It is assumed that the total yield of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events produced in the data is a fixed quantity. This process enters our signal region primarily when a photon from one of the π^0 decays is misidentified as an initial-state radiation photon ($\gamma_{\rm ISR}$). Consequently, two distinct selection efficiencies can be calculated: $\epsilon_{\rm sig}$ (the efficiency for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events to be selected as signal candidates) and $\epsilon_{\rm bkg}$ (the efficiency for the same process to enter the 3π signal region as background).

When $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events are selected as signal, the same analysis procedure as for the signal is applied to construct the 3π invariant mass spectrum, thereby obtaining the distribution of this process in data. Using the ratio of these two efficiencies, $R = \epsilon_{\rm bkg}/\epsilon_{\rm sig}$, the observed $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ signal distribution can be converted into an estimate of its background distribution in the signal region. Specifically:

$$N_{\rm bkg} = R \times N_{\rm sig}^{\rm obs} = \frac{\epsilon_{\rm bkg}}{\epsilon_{\rm sig}} \times N_{\rm sig}^{\rm obs}, \tag{5}$$

where $N_{\rm sig}^{\rm obs}$ is the number of observed $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ signal events in data, and $N_{\rm bkg}$ is the estimated number of background events from this process in the signal region.

This method provides a data-driven conversion from the observed signal distribution to the background distribution through efficiency correction.

When treating $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ as the signal, the selection criteria are designed such that the same event sample also satisfies a loose selection for $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0$. This ensures that the invariant mass of the three-pion system is automatically defined within the latter process.

The combined selection proceeds as follows: events are required to meet the criteria for both processes. The first step applies the full $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ selection:

- $\pi^+\pi^-\pi^0\pi^0$ **Candidate Selection:**
  - Exactly two good charged tracks.
  - nPi0 ≥ 2 & & nGoodShowers ≥ 4 (ensuring at least two π^0s).
  - Application of a total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - The candidate with the minimum $\chi^2_{\rm 4C}$ is chosen.
  - A requirement of $\chi^2_{\rm KF} < 10$ on the kinematic fit.

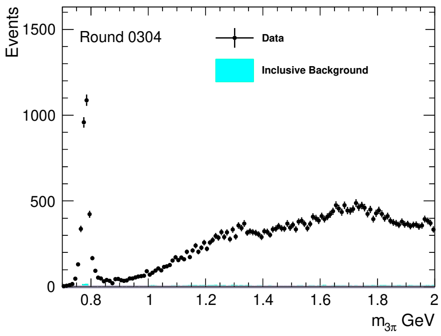
Subsequently, within the same event, a candidate for $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0$ is formed by reinterpreting one of the π^0s as the $\gamma_{\rm ISR}$:

- $\gamma_{\rm ISR}\pi^+\pi^-\pi^0$ **Candidate Selection:**
  - The same two good charged tracks.
  - nPi0 ≥ 1 & & nGoodShowers ≥ 3 (ensuring at least one π^0 and one photon assigned as $\gamma_{\rm ISR}$).
  - Application of the same total 4-momentum constraint to the initial e^+e^- collision (4C kinematic fit).
  - The candidate with the minimum $\chi^2_{\rm 4C}$ is chosen.

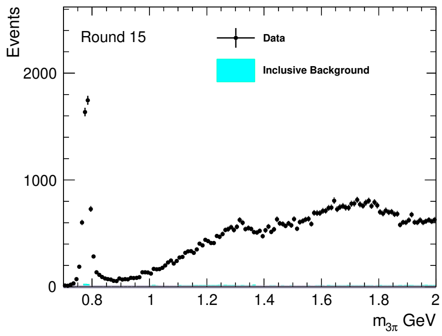
– No additional kinematic fit χ^2 requirement is applied at this stage.

This dual-tag strategy guarantees that the reconstructed 3π system from the $e^+e^- \to \gamma_{\rm ISR}\pi^+\pi^-\pi^0$ hypothesis is kinematically well-defined, and its invariant mass can be reliably calculated for the subsequent analysis.

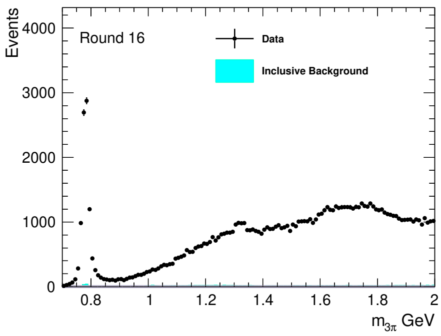
The selection described above results in a highly pure sample of $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events. The invariant mass spectrum of the corresponding three-pion system from this pure sample, which serves as the input template for background estimation, is shown in Fig. 8.



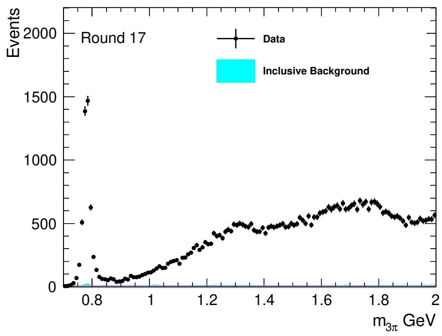
(a) Round 03/04



(b) Round 15



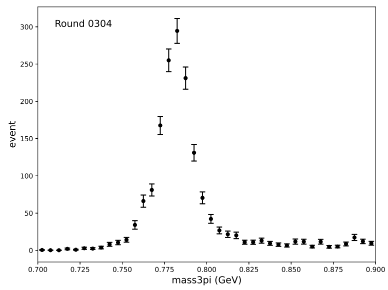
(c) Round 16



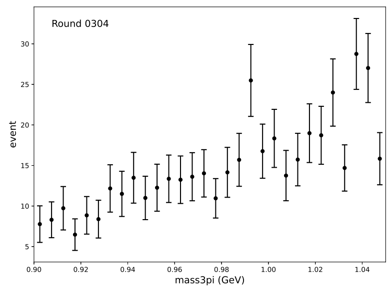
(d) Round 17

Figure 7: The 3π invariant mass spectrum from selected $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events in the (a) Round 03/04, (b) Round 15, (c) Round 16, and (d) Round 17 datasets. These distributions, obtained from data after applying the dual-tag selection criteria for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$, represent the pure signal samples used as templates for background estimation.

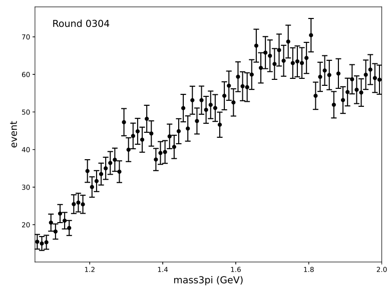
The observed distribution in the data, after applying the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ selection criteria, will be scaled by the efficiency ratio $R = \epsilon_{\rm bkg}/\epsilon_{\rm sig}$ to estimate the final background contribution of this process in the signal region of the measurement channel. As established and shown in Fig. 7, the background contamination within this selected sample is negligible. Therefore, we subtract the inclusive background histogram from the data histogram in a symbolic manner to obtain the pure $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ event yield. The resulting background-subtracted signal distributions in the three 3π invariant mass regions are presented in Fig. **??**. All other backgrounds will be subtracted directly.



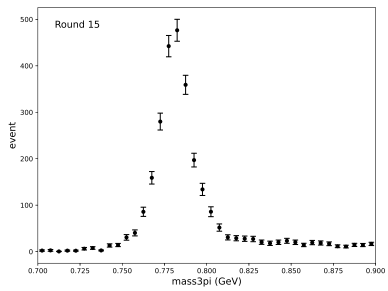
(a) Round 03/04, $\omega(782)$ region



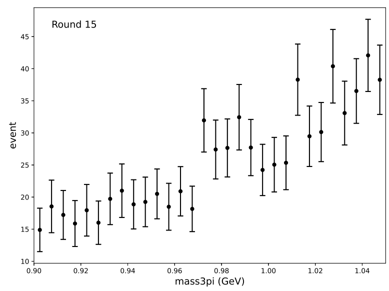
(b) Round 03/04, $\phi(1020)$ region



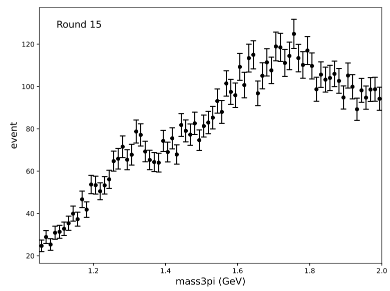
(c) Round 03/04, ω'/ω'' region



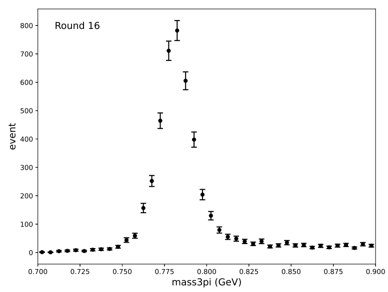
(d) Round 15, $\omega(782)$ region



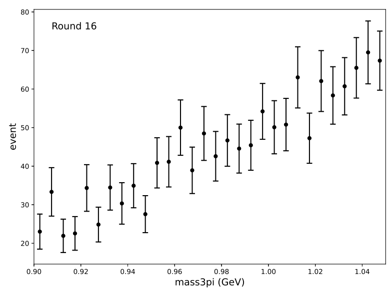
(e) Round 15, $\phi(1020)$ region



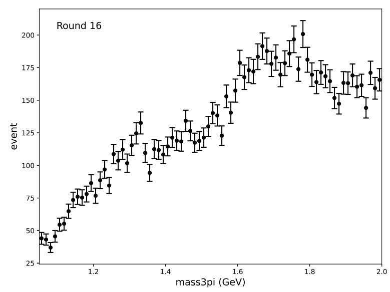
(f) Round 15, ω'/ω'' region



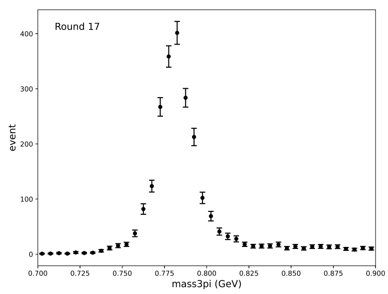
(g) Round 16, $\omega(782)$ region



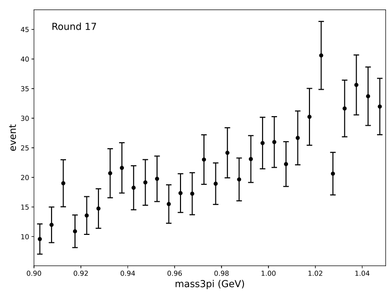
(h) Round 16, $\phi(1020)$ region



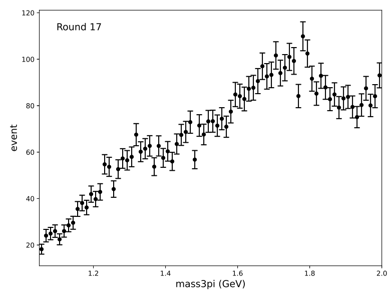
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 8: Estimated background contributions from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ in the signal region after scaling, shown for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right).

5.2 Estimate of Other Backgrounds and Signal Yields

The estimation of the dominant peaking backgrounds from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ has been completed in the previous subsection. These backgrounds exhibit shapes similar to the signal resonance peaks and are therefore categorized as peaking backgrounds.

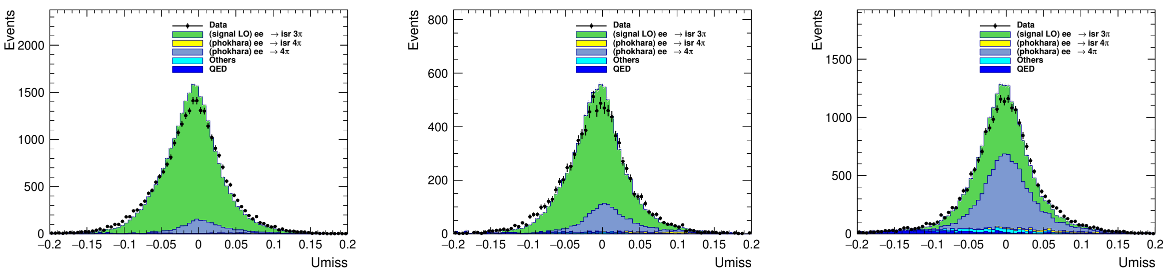
In addition to the peaking component, a non-peaking continuum background component must also be accounted for in the analysis. The primary objective of this study is to measure the cross section for $e^+e^- \to \pi^+\pi^-\pi^0$ via ISR. The ISR photon reduces the effective c.m. energy, allowing the measurement of the signal yield in narrow intervals of the 3π invariant mass ($M_{3\pi}$).

The $M_{3\pi}$ spectrum is divided into three physically motivated regions, each subdivided into multiple bins for detailed study:

- The $\omega(782)$ region (0.7–0.9 GeV$/c^2$), divided into 40 bins.
- The $\phi(1020)$ region (0.9–1.05 GeV$/c^2$), divided into 30 bins.
- The ω'/ω'' region (1.05–2.0 GeV$/c^2$), divided into 76 bins.

For each small $M_{3\pi}$ interval, the signal yield is extracted through an extended unbinned maximum-likelihood fit to the $U_{\rm miss}$ distribution, where $U_{\rm miss}$ is defined as the difference between the missing energy and the magnitude of the missing momentum ($U_{\rm miss} = E_{\rm miss} - |\vec{p}_{\rm miss}|$). The fit model incorporates probability density functions (PDFs) representing the signal and all background components.

To illustrate the composition of $U_{\rm miss}$ distributions across different $M_{3\pi}$ regions, the inclusive MC simulation is examined. Figure 9 shows the $U_{\rm miss}$ distributions for the full $\omega(782)$ region, the full $\phi(1020)$ region, and the full ω'/ω'' region respectively, with the peaking background contribution from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ stacked according to its expected yield. These distributions demonstrate the characteristic features of each mass region that motivate our fitting strategy.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω' and ω'' region

Figure 9: $U_{\rm miss}$ distributions from inclusive MC simulation for three $M_{3\pi}$ regions: (a) the $\omega(782)$ region, (b) the $\phi(1020)$ region, and (c) the higher-mass ω'/ω'' region.

Based on the understanding from the inclusive MC, the PDFs for the fit are constructed as follows:

- **Signal PDF:** Histogram shape from signal MC, convolved with a Gaussian for resolution correction. The $\chi^2_{\rm KF}$ used in the $\chi^2_{\rm KF} < 50$ selection criterion has been corrected for data-MC differences in helix parameters as described in Section 6.5.

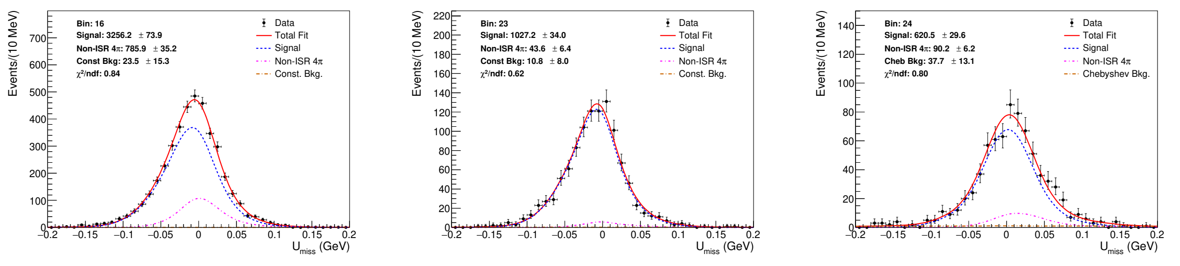
- **Peaking background PDF:** Shape for $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ from dedicated MC, convolved with the same Gaussian as signal. Yield constrained by Gaussian term using data-driven estimate from Section 5.

- **PDF shape sharing across bins:** Limited MC statistics preclude bin-by-bin shape extraction. Instead:

  - For the $\omega(782)$ and $\phi(1020)$ regions, where the $M_{3\pi}$ bin width is 5 MeV/c^2, the PDF shapes are extracted from MC simulation every five consecutive bins, corresponding to 25 MeV/c^2 intervals. All five bins within each such interval share the same PDF shape, under the assumption that the $U_{\rm miss}$ distribution varies slowly over this mass range.
  - For the ω'/ω'' region, where the $M_{3\pi}$ bin width is 12.5 MeV/c^2, the PDF shapes are also extracted every 25 MeV/c^2. Each extracted shape is therefore shared by two adjacent bins, balancing the need for shape precision against the limited MC statistics in this higher-mass region.

- **Continuum background PDF:** All remaining minor background processes are collectively described by a polynomial function. A first-order polynomial (linear function) is used for the $\omega(782)$ and $\phi(1020)$ regions, while a second-order Chebyshev polynomial is employed for the ω'/ω'' region, as suggested by the shape of the inclusive MC distribution in Fig. 9c.

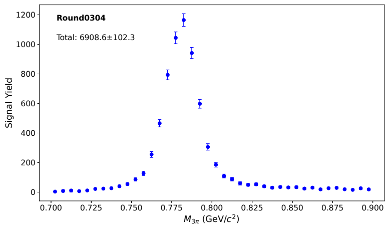
The fit procedure is validated by examining the quality of the fits to data in representative $M_{3\pi}$ bins across the three mass regions, using Round 16 as an illustrative example. Figure 10 presents typical fit results for three selected bins, one from each region, demonstrating the ability of the fit model to accurately describe the data. The signal component, peaking background from $\pi^+\pi^-\pi^0\pi^0$, and polynomial continuum background are clearly distinguished, and the total PDF provides an excellent description of the data distribution in each case.



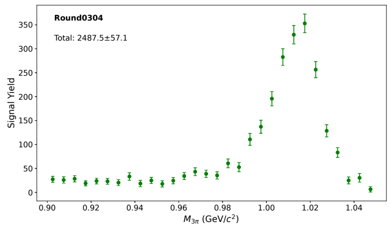
(a) Example bin in the $\omega(782)$ region (b) Example bin in the $\phi(1020)$ region (c) Example bin in the ω'/ω'' region

Figure 10: Typical $U_{\rm miss}$ fit results for representative $M_{3\pi}$ bins from each of the three mass regions, shown for Round 16 data.

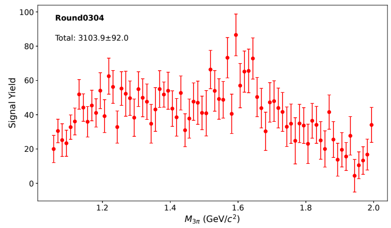
Applying this fitting procedure to all $M_{3\pi}$ bins yields the extracted signal yields across the entire mass spectrum. The same fit is performed independently for four data-taking periods: Round 03/04, Round 15, Round 16, and Round 17. Figure 11 presents the measured signal yields as a function of $M_{3\pi}$ for all four periods, organized by the three mass regions. In each panel, the $\omega(782)$ resonance is clearly visible as a prominent peak around 0.78 GeV/c^2, while the $\phi(1020)$ resonance appears as a distinct but smaller peak near 1.02 GeV/c^2. In the higher mass regions, broad structures corresponding to the ω' and ω'' excitations are observed, with significantly reduced yields due to the decreasing cross section and lower ISR photon probability at larger momentum transfers. The results show good consistency across the four data-taking periods, with statistical uncertainties represented by the error bars.



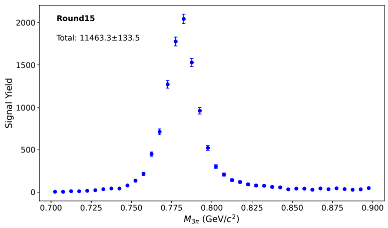
(a) Round 03/04, $\omega(782)$ region



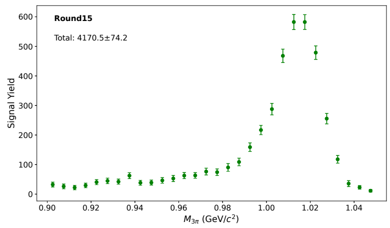
(b) Round 03/04, $\phi(1020)$ region



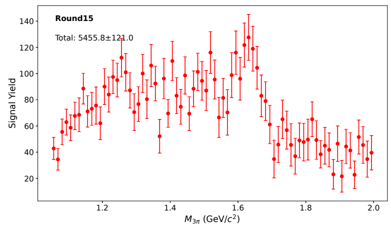
(c) Round 03/04, ω'/ω'' region



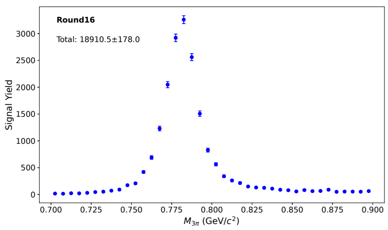
(d) Round 15, $\omega(782)$ region



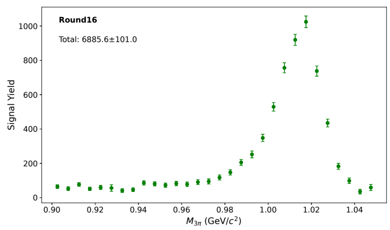
(e) Round 15, $\phi(1020)$ region



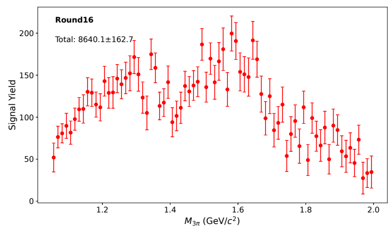
(f) Round 15, ω'/ω'' region



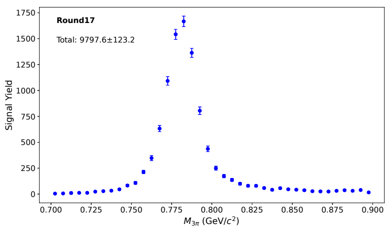
(g) Round 16, $\omega(782)$ region



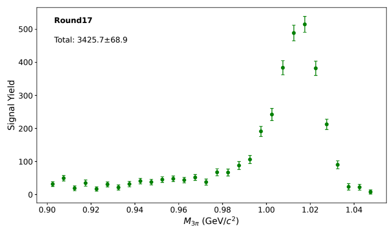
(h) Round 16, $\phi(1020)$ region



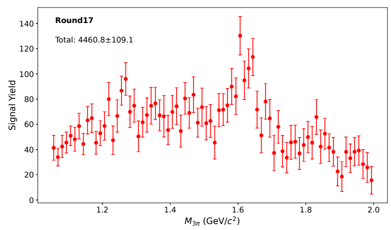
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 11: Extracted signal yields as a function of $M_{3\pi}$ for the three mass regions, shown separately for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). Error bars represent the statistical uncertainties from the fits.

6 Correction of Signal Efficiency

To extract the dressed cross section $\sigma_{3\pi}^{\mathrm{dress}}(m)$, we need the generator-level yield $\left(\frac{dN}{dm}\right)^{\mathrm{gen}}$, which is related to the cross section within each mass interval by:

$$\left(\frac{dN}{dm}\right)^{\mathrm{gen}} = \sigma_{3\pi}^{\mathrm{dress}}(m) \cdot \frac{dL}{dm} \cdot \varepsilon \cdot \mathcal{B}(\pi^0 \to \gamma\gamma) \tag{6}$$

where $\frac{dL}{dm}$ is the integrated luminosity in that interval, and $\mathcal{B}(\pi^0 \to \gamma\gamma)$ is the branching fraction of the $\pi^0 \to \gamma\gamma$ decay.

From data, we have obtained the observed signal yield $\left(\frac{dN}{dm}\right)_i^{\mathrm{obs}}$ in each mass interval by fitting the U_{miss} distribution, as shown in Fig. 11. Due to detector effects, events can migrate between different mass intervals, so these observed yields do not directly give the generator-level yields in Eq. (6). The migration is described by the response matrix P_{ij}, which relates the observed yield in bin i to the generator-level yield in bin j:

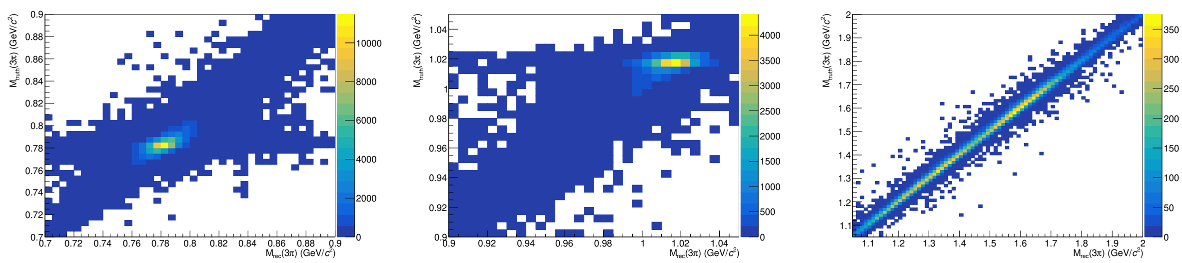
$$\left(\frac{dN}{dm}\right)_i^{\mathrm{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\mathrm{gen}} \tag{7}$$

The generator-level yields $\left(\frac{dN}{dm}\right)_j^{\mathrm{gen}}$ are therefore obtained by solving Eq. (21) for $\left(\frac{dN}{dm}\right)_j^{\mathrm{gen}}$ given the observed yields $\left(\frac{dN}{dm}\right)_i^{\mathrm{obs}}$ and the response matrix P_{ij}. This numerical procedure is referred to as unfolding.

Let A_{ij} denote the number of MC events that are generated in true bin j and reconstructed in bin i, where i and j index the observed and generator-level mass bins, respectively. The migration matrices A_{ij} for three representative mass regions are shown in Fig. 12, illustrating how events generated in a given true bin are distributed across observed bins due to detector effects. This matrix is then normalized column-wise to obtain the response matrix used in Eq. (21):

$$P_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_d} A_{kj}}, \qquad \sum_{i=1}^{n_d} P_{ij} = 1, \tag{8}$$

where n_d is the number of reconstructed bins.



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω'/ω'' region

Figure 12: Response matrices A_{ij} for three representative mass regions, showing the migration of events between generator-level and reconstructed $M_{3\pi}$ bins. The diagonal structure indicates good resolution, while off-diagonal entries reflect the effects of finite detector resolution and acceptance.

Since the unfolding procedure relies on the response matrix derived from MC simulation, any discrepancies between data and MC can lead to biased results. Therefore, the MC sample used to construct the response matrix must be corrected to ensure that it accurately reflects the data. These discrepancies arise from various sources: the tracking efficiency of charged particles, PID performance, the reconstruction efficiency of neutral π^0 mesons, the detection efficiency of ISR photons, differences in the production rate of additional ISR photons (NLO) between the Phokhara generator and data, and the discrepancy in the χ^2_{4C} distribution between data and MC. The following subsections address each of these corrections in detail.

6.1 Tracking and PID efficiencies

The tracking and PID efficiencies for charged pions are adopted from the dedicated study in Ref. [14], where they are measured using $D^0\bar{D}^0$ and D^+D^- events. The kinematic coverage of that study matches the momentum range of charged pions in this analysis. For each momentum bin, the correction factors $c_{\mathrm{trk}}(p)$ and $c_{\mathrm{pid}}(p)$ are derived separately for tracking and PID.

6.2 The π^0 reconstruction efficiency

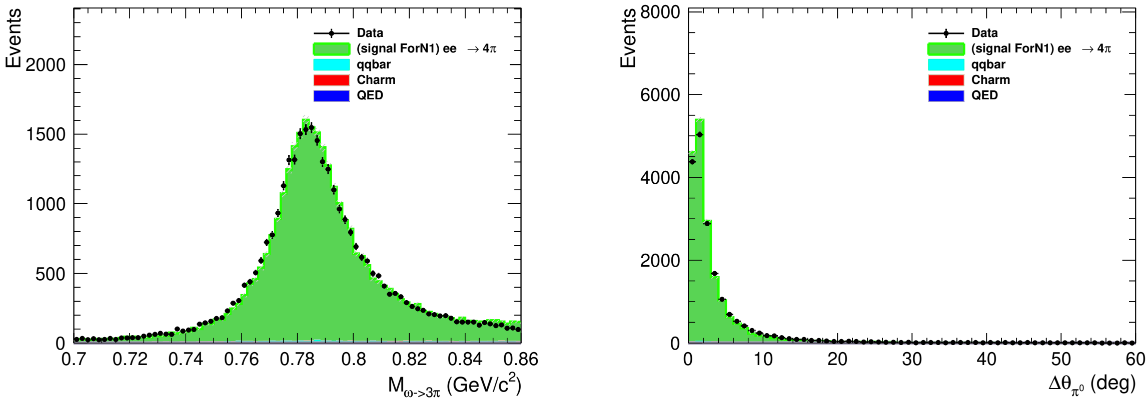
The reconstruction efficiency for π^0 mesons is studied using a dedicated control sample of $e^+e^- \to \omega\pi^0_{\mathrm{tag}}$ events at $\sqrt{s} = 3.773\,\mathrm{GeV}$, with $\omega \to \pi^+\pi^-\pi^0_{\mathrm{pred}}$. This process provides a clean environment to study π^0 efficiency due to the distinct kinematic properties of the two neutral pions: the π^0_{tag} carries approximately half of the center-of-mass momentum, while the π^0_{pred} from ω decay has a momentum spectrum ranging from 0 to $\sqrt{s}/2$, making it well suited for a momentum-dependent efficiency study.

The measurement is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) using the 3.773 GeV data sample. The selection criteria are as follows:

- The selection requirements for charged pions and π^0 reconstruction are identical to those used in the signal channel $e^+e^- \to \pi^+\pi^-\pi^0$.
- Events are required to have at least one reconstructed π^0 candidate.
- In events with multiple π^0 candidates, the two with the best fit quality from 1C kinematic fitting are retained for further analysis.
- The π^0_{tag} is identified as the π^0 candidate whose energy is closest to half the center-of-mass energy ($\sqrt{s}/2$), with a requirement that the difference be within 0.3 GeV.
- A 1C kinematic fit is then performed: the recoil momentum of the $\pi^+\pi^-\pi^0_{\mathrm{tag}}$ system against the initial e^+e^- system is calculated in the center-of-mass frame; this recoil momentum is constrained to the π^0 mass, corresponding to the missing π^0_{pred}; subsequently, the $\pi^+\pi^-$ system together with the π^0_{tag} is constrained to the center-of-mass system.

The 1C kinematic fit is required to satisfy $\chi^2 < 15$. After the fit, the four-momentum of the π^0_{pred} is updated. The invariant mass of the $\pi^+\pi^-\pi^0_{\mathrm{pred}}$ system is then calculated, and events are required to have this invariant mass within the ω meson mass window of 0.7–0.86 GeV.

Before applying the momentum-dependent efficiency study, we first examine the overall $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution without requiring a matching reconstructed π^0 in the predicted direction. Figure 13a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.



(a) (b)

Figure 13: (a) Full $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the ω mass window used for event selection. (b) Angular separation $\Delta\theta$ between the predicted $\pi^0_{\rm pred}$ direction and additional reconstructed π^0 candidates. The vertical dashed line indicates the 20° matching cone used in this analysis.

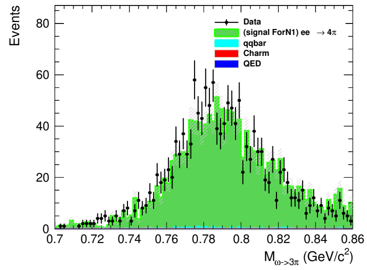
To associate a reconstructed π^0 with the predicted $\pi^0_{\rm pred}$, we examine the angular separation $\Delta\theta$ between the direction of the predicted $\pi^0_{\rm pred}$ and any additional reconstructed π^0 candidate (excluding the $\pi^0_{\rm tag}$). Figure 13b shows the distribution of this angular separation for events where an additional π^0 is found. A clear peak at small angles is observed, corresponding to correctly matched π^0 candidates. Based on this distribution, we define a matching cone of 20°: if an additional reconstructed π^0 lies within 20° of the predicted direction, it is considered as the reconstructed $\pi^0_{\rm pred}$.

For each momentum bin, Fig. 14 shows the $\pi^+\pi^-\pi^0_{\rm pred}$ invariant mass distribution for events where a matching reconstructed π^0 is found within the 20° cone. The distributions for $p_{\pi^0} < 0.2$ GeV, $0.2 \le p_{\pi^0} < 0.4$ GeV, $0.4 \le p_{\pi^0} < 0.6$ GeV, $0.6 \le p_{\pi^0} < 0.8$ GeV, $0.8 \le p_{\pi^0} < 1.0$ GeV, $1.0 \le p_{\pi^0} < 1.2$ GeV, $1.2 \le p_{\pi^0} < 1.4$ GeV, and $p_{\pi^0} \ge 1.4$ GeV all show clean signal peaks, demonstrating that the matching procedure is effective for the entire momentum range.

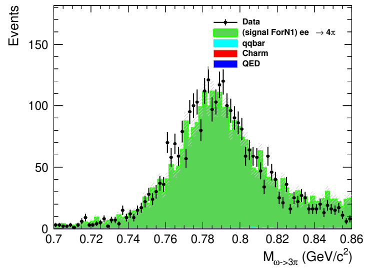
Given the negligible background in the signal region, the π^0 reconstruction efficiency can be directly determined by counting events. For each momentum bin, we define:

- $N^{\rm total}_{\rm data}(p)$: number of data events in the ω mass window, regardless of whether a matching π^0 is found.

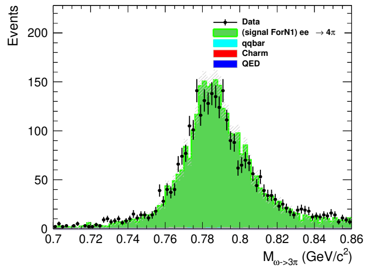
- $N^{\rm matched}_{\rm data}(p)$: number of data events in the ω mass window with a matching reconstructed π^0 within 20° of the predicted direction.



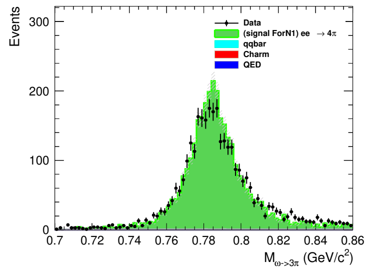
(a) Bin 0: $p < 0.2\,\mathrm{GeV}$



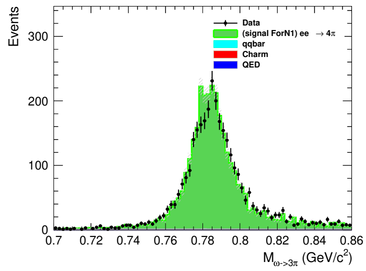
(b) Bin 1: $0.2 \leq p < 0.4\,\mathrm{GeV}$



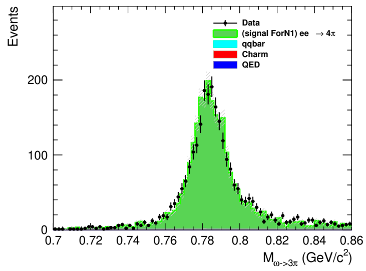
(c) Bin 2: $0.4 \leq p < 0.6\,\mathrm{GeV}$



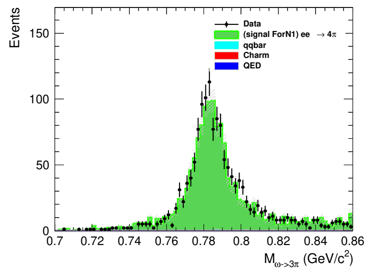
(d) Bin 3: $0.6 \leq p < 0.8\,\mathrm{GeV}$



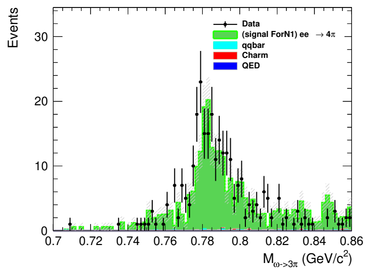
(e) Bin 4: $0.8 \leq p < 1.0\,\mathrm{GeV}$



(f) Bin 5: $1.0 \leq p < 1.2\,\mathrm{GeV}$



(g) Bin 6: $1.2 \leq p < 1.4\,\mathrm{GeV}$



(h) Bin 7: $p \geq 1.4\,\mathrm{GeV}$

Figure 14: $\pi^+\pi^-\pi^0_{\mathrm{pred}}$ invariant mass distributions for events with a matching reconstructed π^0 within 20°, shown for each π^0_{pred} momentum bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the ω mass window used for event selection.

- $N_{\mathrm{MC}}^{\mathrm{total}}(p)$ and $N_{\mathrm{MC}}^{\mathrm{matched}}(p)$: corresponding quantities for MC.

The efficiencies in data and MC are then computed as:

$$\varepsilon_{\mathrm{data}}(p) = \frac{N_{\mathrm{data}}^{\mathrm{matched}}(p)}{N_{\mathrm{data}}^{\mathrm{total}}(p)}, \quad \varepsilon_{\mathrm{MC}}(p) = \frac{N_{\mathrm{MC}}^{\mathrm{matched}}(p)}{N_{\mathrm{MC}}^{\mathrm{total}}(p)}. \tag{9}$$

The correction factor for π^0 reconstruction is then derived as the ratio of data efficiency to MC efficiency:

$$c_{\pi^0}(p) = \frac{\varepsilon_{\mathrm{data}}(p)}{\varepsilon_{\mathrm{MC}}(p)}. \tag{10}$$

The calculated efficiencies and correction factors are presented in Table 4. Each entry is shown as central value ± its statistical uncertainty.

Table 4: The π^0 reconstruction efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different π^0 momentum intervals.

p_{π^0} (GeV)	$\varepsilon_{\mathrm{data}}$	$\varepsilon_{\mathrm{MC}}$	c_{π^0}
< 0.2	0.3892 ± 0.0080	0.3958 ± 0.0064	0.9833 ± 0.0254
0.2 – 0.4	0.4579 ± 0.0061	0.4782 ± 0.0050	0.9575 ± 0.0168
0.4 – 0.6	0.5442 ± 0.0067	0.5636 ± 0.0059	0.9656 ± 0.0163
0.6 – 0.8	0.6265 ± 0.0068	0.6471 ± 0.0060	0.9682 ± 0.0150
0.8 – 1.0	0.6946 ± 0.0068	0.7124 ± 0.0060	0.9750 ± 0.0142
1.0 – 1.2	0.7540 ± 0.0074	0.7785 ± 0.0064	0.9685 ± 0.0139
1.2 – 1.4	0.8089 ± 0.0094	0.8161 ± 0.0082	0.9912 ± 0.0167
≥ 1.4	0.8017 ± 0.0215	0.8574 ± 0.0155	0.9350 ± 0.0283

6.3 The ISR photon detection efficiency

The ISR photon detection efficiency is studied using the same control sample of $e^+e^- \to \omega\pi^0$ events at $\sqrt{s} = 3.773\,\mathrm{GeV}$, with $\omega \to \pi^+\pi^-\pi^0$. Unlike the previous π^0 efficiency study which tagged one π^0 and studied the other from ω decay, here we identify one π^0 as coming from ω decay and study the photon detection efficiency using the two photons from the decay of the remaining π^0. This remaining π^0, which balances the ω in the event, carries approximately half of the center-of-mass energy ($\sqrt{s}/2$).

The measurement is performed independently for each data-taking period using the 3.773 GeV data sample. The selection criteria are as follows:

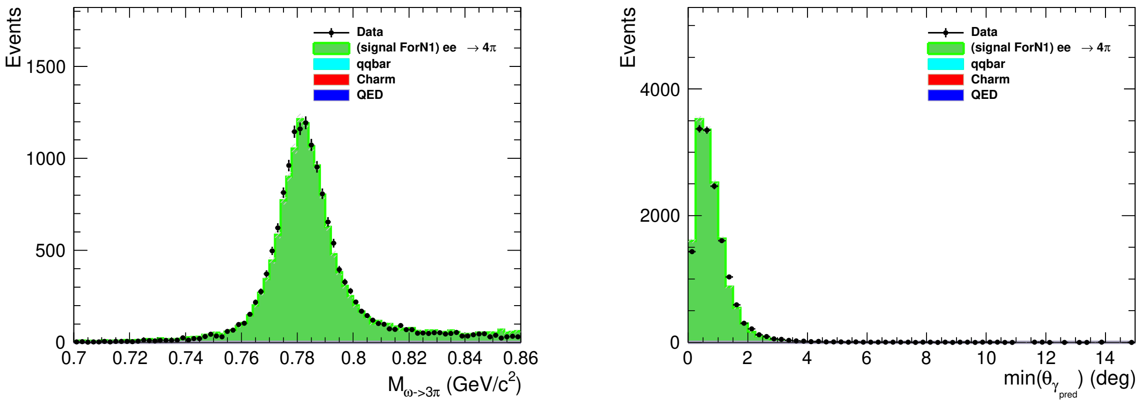
- The selection requirements for charged pions are identical to those used in the signal channel $e^+e^- \to \pi^+\pi^-\pi^0$.
- Events are required to have at least one reconstructed π^0 candidate.

- For each reconstructed π^0 candidate, the invariant mass of the $\pi^+\pi^-$ system combined with this π^0 is calculated. Among all π^0 candidates, the one that yields an invariant mass closest to the nominal ω mass and within ± 0.3 GeV is identified as originating from the ω decay (π^0_ω).
- The other π^0 in the event (denoted as π^0_{bal}) balances the ω system and has a characteristic energy of approximately $\sqrt{s}/2$. It decays into two photons: $\pi^0_{\text{bal}} \to \gamma\gamma$. Note that this π^0_{bal} itself is not required to be reconstructed; we only study its decay photons, and its existence is inferred from the ω system and momentum conservation.
- All remaining photon candidates in the event (excluding those used in π^0_ω reconstruction) are considered as potential tagged photons. For each such photon candidate, we treat it as the tagged photon (γ_{tag}) and perform a 2C kinematic fit where the other photon (γ_{pred}) is inferred via momentum conservation.
- A 2C kinematic fit is performed for each photon candidate with the following constraints:
  - The invariant mass of the missing momentum (corresponding to γ_{pred}) is constrained to zero (photon mass).
  - The invariant mass of the tagged photon and the predicted photon is constrained to the π^0 mass.
  - The total four-momentum of the event is constrained to the center-of-mass system ($\sqrt{s} = 3.773$ GeV).
- Entries with $\chi^2 < 5$ from the kinematic fit are classified as good entries. Events are required to have at least one good entry.
- For events with multiple good entries, the two with the smallest χ^2 are retained. From these two, the entry with the smaller tagged photon energy (correspondingly larger predicted photon energy) is selected for further analysis. This selection ensures that the predicted photon energy lies in the region of interest above 1 GeV.
- The selected entry is required to satisfy the kinematic fit quality $\chi^2 < 5$.

Before applying the energy-dependent efficiency study, we first examine the overall $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution without requiring a matching reconstructed photon in the predicted direction. Figure 15a shows this distribution for all events passing the selection criteria, with data, MC, and estimated background contributions overlaid. The signal region 0.7–0.86 GeV exhibits a clean peak with negligible background, confirming the purity of the control sample.

To associate a reconstructed photon with the predicted photon γ_{pred}, we examine the angular separation $\Delta\theta$ between the direction of the predicted photon and any reconstructed photon candidate (excluding those used in π^0_ω reconstruction). Figure 15b shows the distribution of this angular separation for events where a photon is found. A clear peak at small angles is observed, corresponding to correctly matched photon candidates. Based on this distribution, we define a matching cone of 4○: if a reconstructed photon lies within 4° of the predicted direction, it is considered as the reconstructed γ_{pred}.

For each energy bin, Fig. 16 shows the $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution for events where a matching reconstructed photon is found within the 4° cone. The distributions for $1.0 \le E_\gamma < 1.1$ GeV,



(a) (b)

Figure 15: (a) Full $\pi^+\pi^-\pi^0_\omega$ invariant mass distribution for all selected events, showing data (points), MC (green), and estimated background (other colors). The vertical dashed lines indicate the ω mass window used for event selection. (b) Angular separation $\Delta\theta$ between the predicted photon γ_{pred} direction and reconstructed photon candidates. The vertical dashed line indicates the 4° matching cone used in this analysis.

$1.1 \leq E_\gamma < 1.2$ GeV, $1.2 \leq E_\gamma < 1.3$ GeV, $1.3 \leq E_\gamma < 1.4$ GeV, $1.4 \leq E_\gamma < 1.5$ GeV, $1.5 \leq E_\gamma < 1.6$ GeV, $1.6 \leq E_\gamma < 1.7$ GeV, and $E_\gamma \geq 1.8$ GeV all show clean signal peaks, demonstrating that the matching procedure is effective for the entire energy range.

Given the negligible background in the signal region, the photon detection efficiency can be directly determined by counting events. For each energy bin, we define:

- $N^{\text{total}}_{\text{data}}(E)$: number of data events in the ω mass window, regardless of whether a matching photon is found.
- $N^{\text{matched}}_{\text{data}}(E)$: number of data events in the ω mass window with a matching reconstructed photon within 4° of the predicted direction.
- $N^{\text{total}}_{\text{MC}}(E)$ and $N^{\text{matched}}_{\text{MC}}(E)$: corresponding quantities for MC.

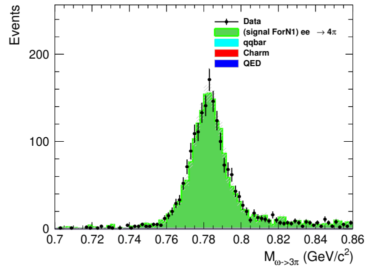
The efficiencies in data and MC are then computed as:

$$\varepsilon_{\text{data}}(E) = \frac{N^{\text{matched}}_{\text{data}}(E)}{N^{\text{total}}_{\text{data}}(E)}, \quad \varepsilon_{\text{MC}}(E) = \frac{N^{\text{matched}}_{\text{MC}}(E)}{N^{\text{total}}_{\text{MC}}(E)}. \tag{11}$$

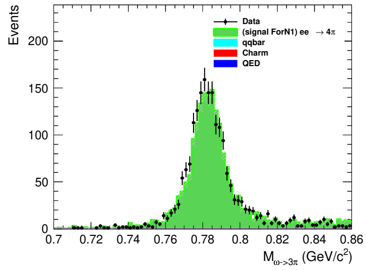
The correction factor for photon detection is then derived as the ratio of data efficiency to MC efficiency:

$$c_\gamma(E) = \frac{\varepsilon_{\text{data}}(E)}{\varepsilon_{\text{MC}}(E)}. \tag{12}$$

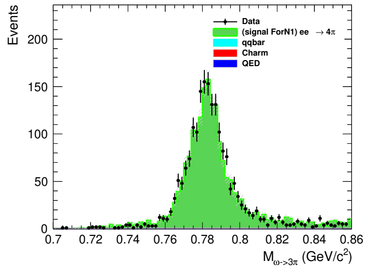
The calculated efficiencies and correction factors are presented in Table 5. Each entry is shown as central value ± its statistical uncertainty.



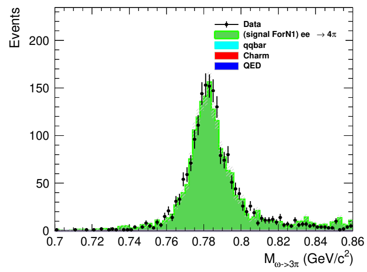
(a) Bin 1: $1.0 \leq E < 1.1$ GeV



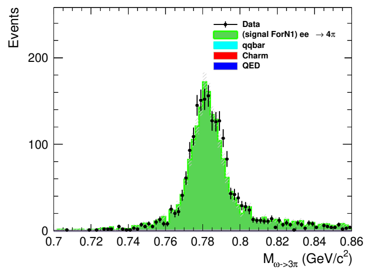
(b) Bin 2: $1.1 \leq E < 1.2$ GeV



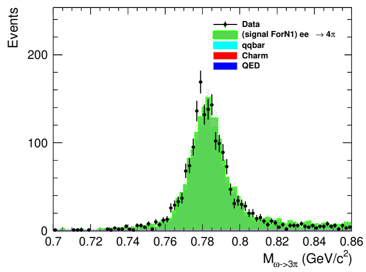
(c) Bin 3: $1.2 \leq E < 1.3$ GeV



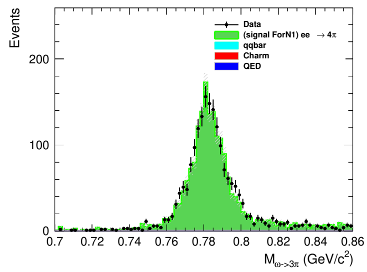
(d) Bin 4: $1.3 \leq E < 1.4$ GeV



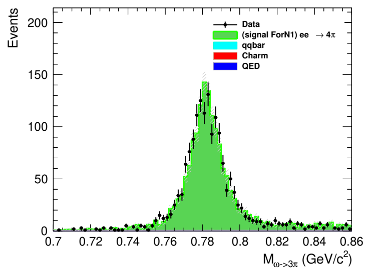
(e) Bin 5: $1.4 \leq E < 1.5$ GeV



(f) Bin 6: $1.5 \leq E < 1.6$ GeV



(g) Bin 7: $1.6 \leq E < 1.7$ GeV



(h) Bin 8: $E \geq 1.8$ GeV

Figure 16: $\pi^+\pi^-\pi^0_\omega$ invariant mass distributions for events with a matching reconstructed photon within 4°, shown for each predicted photon energy bin. Data (points), MC (green), and estimated background (other colors) are overlaid. The vertical dashed lines indicate the ω mass window used for event selection.

Table 5: The photon detection efficiencies in data and MC, and the derived correction factors with their statistical uncertainties for different photon energy intervals. The correction factor uncertainties are calculated using error propagation.

E_γ (GeV)	$\varepsilon_{\text{data}}$	ε_{MC}	c_γ
1.0 – 1.1	0.9744 ± 0.0036	0.9791 ± 0.0027	0.9952 ± 0.0045
1.1 – 1.2	0.9726 ± 0.0038	0.9799 ± 0.0026	0.9926 ± 0.0046
1.2 – 1.3	0.9755 ± 0.0035	0.9798 ± 0.0026	0.9956 ± 0.0044
1.3 – 1.4	0.9724 ± 0.0037	0.9799 ± 0.0026	0.9923 ± 0.0045
1.4 – 1.5	0.9748 ± 0.0035	0.9821 ± 0.0025	0.9926 ± 0.0043
1.5 – 1.6	0.9796 ± 0.0032	0.9873 ± 0.0021	0.9922 ± 0.0039
1.6 – 1.7	0.9690 ± 0.0040	0.9846 ± 0.0024	0.9842 ± 0.0047
≥ 1.8	0.9740 ± 0.0040	0.9824 ± 0.0028	0.9915 ± 0.0049

6.4 PHOKHARA Extra Photon Fraction

A BaBar study of the PHOKHARA generator used for signal simulation found a 20% excess of events with an additional energetic ISR photon along the beam line, relative to data in their $e^+e^- \to \pi^+\pi^-$ process [15]. The ISR-based analysis at BaBar is not impacted as it relied on a different generator, but given that our signal process $e^+e^- \to \pi^+\pi^-\pi^0$ is also simulated using PHOKHARA and involves ISR photon emission, this discrepancy warrants a cross-check in our analysis.

To investigate this effect in our channel, we examine the fraction of events with an additional ISR photon directly within our signal process $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\text{ISR}}$. This corresponds to studying events of the type $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\text{ISR}}\gamma_{\text{NLO}}$, where two ISR photons are present in the final state. NNLO contributions are not considered, as their production probability is significantly lower and the additional photons are typically too soft to be reliably identified given our detector resolution and event statistics. Moreover, the limited data sample size does not provide sufficient statistical power to meaningfully extract an NNLO contribution.

In this study, we rely on the ISR photon identified through the 4C kinematic fit procedure described in Sec. 4, which selects the optimal candidate event and tags the leading order (LO) ISR photon with high confidence. Based on this tagged photon, we proceed to investigate the presence of additional photons in the event. Due to the finite coverage of the BESIII electromagnetic calorimeter, the study of NLO photons is divided into two independent categories based on their angular distribution. As this study focuses on the intrinsic discrepancy between data and the PHOKHARA generator, the potential differences between different data-taking rounds are not considered. We perform this investigation using the Round16 dataset and assume the observed effect is representative and can be generalized to the other rounds.

6.4.1 Large Angle NLO Photons

NLO photons emitted at large angles fall within the detector acceptance and can be directly reconstructed from electromagnetic showers. Events containing a large-angle NLO photon are selected by first inverting the nominal signal criterion described in Sec. 4: we require the 4C kinematic fit under the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ hypothesis to have $\chi^2 > 50$, as events passing the $\chi^2 < 50$ signal selection are, by construction, depleted of additional photons. From this complementary sample, we identify the NLO photon candidate as the extra reconstructed photon that minimizes the χ^2 of a subsequent 4C kinematic fit to the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}\gamma_{\rm NLO}$ hypothesis, which explicitly includes both ISR photons in the final state. The kinematic fit is performed in the laboratory frame, constraining the total four-momentum of all final state particles (π^+, π^-, π^0, the LO ISR photon, and the NLO LA photon) to the initial e^+e^- center-of-mass energy. To ensure a well-reconstructed NLO topology, we further require this 4C fit to satisfy $\chi^2 < 50$. The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.

To suppress background contributions from events where the two ISR photons originate from π^0 or η decays, we examine the invariant mass squared distribution of the LO and NLO LA photon pair, $M^2(\gamma_{\rm ISR}\gamma_{\rm NLO})$. Clear peaks corresponding to the π^0 and η mesons are observed, as shown in Fig. 17. Accordingly, we apply veto requirements of $M^2(\gamma_{\rm ISR}\gamma_{\rm NLO}) < 0.05$ GeV$^2/c^4$ to exclude the π^0 region, and $0.23 < M^2(\gamma_{\rm ISR}\gamma_{\rm NLO}) < 0.33$ GeV$^2/c^4$ to exclude the η region.

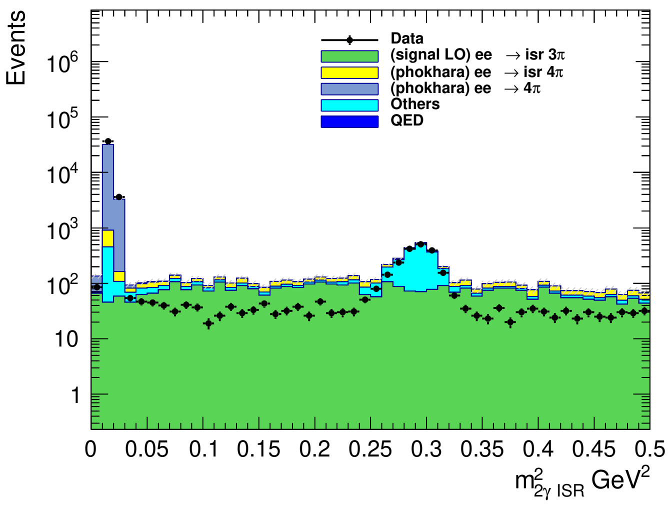
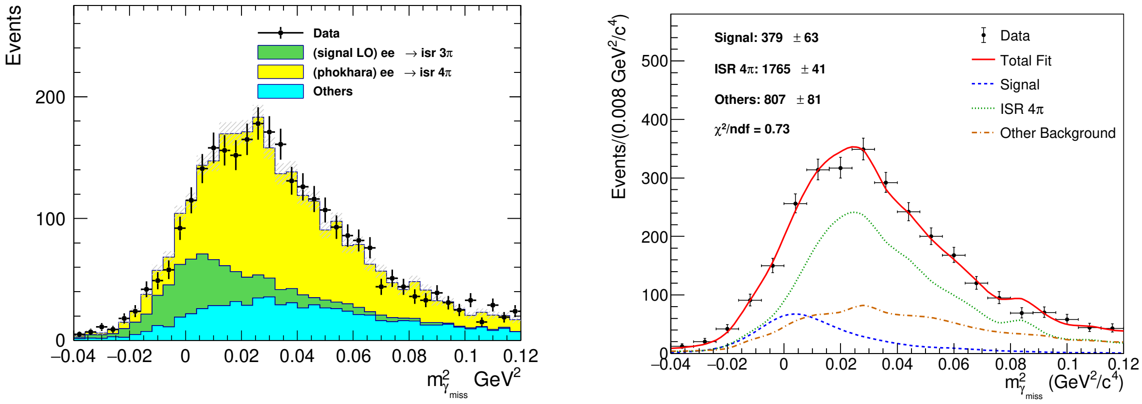


Figure 17: Invariant mass squared distribution of the two ISR photons.

To further validate the selection and extract the NLO LA photon yield, we examine the invariant mass squared of the missing system: the total laboratory four-momentum minus the sum of the π^+, π^-, π^0, and LO ISR photon four-momenta. For signal events, the remaining system corresponds exactly to the NLO ISR photon, and this quantity should therefore peak at zero. Figure 18a shows this distribution for events passing the inverted single-ISR selection ($\chi^2 > 50$) with an additional photon candidate.

To extract the signal yield, we perform a fit to this distribution, as shown in Figure 18b:

- The signal shape is taken from simulated $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}\gamma_{\rm NLO}$ events and convolved with a Gaussian to account for possible resolution differences between data and MC. The signal yield is



(a) (b)

Figure 18: (a) Invariant mass squared of the missing system (total momentum minus $\pi^+\pi^-\pi^0\gamma_{\rm ISR}$). (b) Fit to the distribution.

left free in the fit. The Gaussian resolution parameters are extracted from the fit.

- The dominant background from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events is modeled using the shape extracted from simulation. Its yield is constrained using a Gaussian penalty term based on the expected number of events, calculated from the luminosity, the cross-section, and the selection efficiency estimated from MC, with a Poisson error incorporated to account for statistical fluctuations.

- All other remaining background contributions are modeled using shapes extracted from MC samples, with their yields left free in the fit.

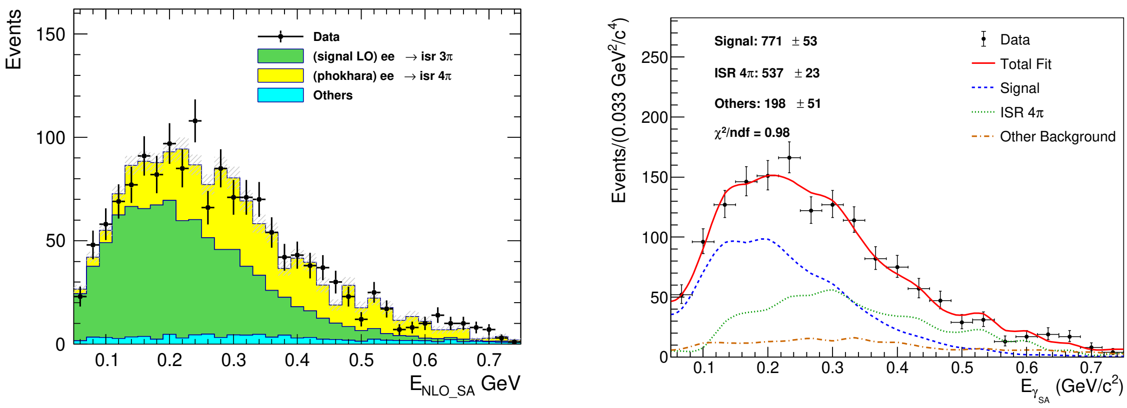
The fitted NLO LA signal yield is 379 ± 63. The large uncertainty is due to the low production rate of these events and the significant photon background, which makes it difficult to improve the significance.

6.4.2 Small Angle NLO Photons

NLO photons emitted at very small angles along the beam direction escape detection, and their kinematic properties must be inferred indirectly. The momentum of this undetected photon is calculated from the missing momentum, with the constraint that its transverse components are zero in the center-of-mass frame ($p_x^{\rm CM} = p_y^{\rm CM} = 0$). It is important to note that the beam direction in the laboratory frame is not aligned with the definition of transverse momentum in this context; therefore, we first boost the event to the center-of-mass frame, enforce the transverse momentum condition, and then boost back to the laboratory frame to obtain the four-momentum of the SA photon. The particle is further constrained to have zero mass (photon identity). A 3C kinematic fit is then performed in the laboratory frame, including the detected particles π^+, π^-, π^0, and the LO ISR photon, together with the inferred SA photon, to enforce overall energy and momentum conservation. The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.

Events containing a small-angle NLO photon are selected by first inverting the nominal signal criterion described in Sec. 4: we require the 4C kinematic fit under the $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\rm ISR}$ hypothesis to have $\chi^2 > 50$. From this complementary sample, we infer the SA photon candidate using the missing momentum technique described above. To ensure a well-constrained NLO topology, we require the 3C kinematic fit to satisfy $\chi^2 < 5$. Additionally, we examine the angular distribution of the inferred SA photon in the center-of-mass frame and require $|\cos\theta^{\mathrm{CM}}_{\gamma_{\mathrm{NLO}}}| > 0.99$, consistent with the small-angle region where photons escape detection.

To further validate the selection and extract the NLO SA photon yield, we examine the energy distribution of the inferred SA photon, $E^{\mathrm{lab}}_{\gamma_{\mathrm{NLO}}}$. For signal events, this energy should exhibit a characteristic spectrum determined by the NLO radiation process. Figure 19 shows this distribution for events passing the inverted single-ISR selection and the additional SA photon requirements.



(a) (b)

Figure 19: (a) Laboratory energy distribution of the inferred SA photon candidate. (b) Fit to the distribution.

To extract the signal yield, we perform a fit to this distribution, as shown in Figure 19b:

- The signal shape is taken from simulated $e^+e^- \to \pi^+\pi^-\pi^0\gamma_{\mathrm{ISR}}\gamma_{\mathrm{NLO}}$ events and convolved with a Gaussian to account for possible resolution differences between data and MC. The signal yield is left free in the fit. The Gaussian resolution parameters are extracted from the fit.

- The dominant background from $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ events is modeled using the shape extracted from simulation. Its yield is constrained using a Gaussian penalty term based on the expected number of events, calculated from the luminosity, the cross-section, and the selection efficiency estimated from MC, with a Poisson error incorporated to account for statistical fluctuations.

- All other remaining background contributions are modeled using shapes extracted from MC samples, with their yields left free in the fit.

The fitted NLO SA signal yield is 771 ± 53.

6.4.3 Data-MC Comparison of NLO Photon Fractions

Following the selection procedures described above for both LA and SA categories, we obtain samples of events containing an identified NLO ISR photon in data. To compare the production rates between data and the PHOKHARA generator, we need to account for the overall selection efficiency that encompasses both detector effects and reconstruction requirements.

Using the PHOKHARA MC sample, we first classify events at generator level according to the criteria defined as follows, with an additional requirement on the invariant mass of the three pions consistent with our analysis region:

- The invariant mass $M(\pi^+\pi^-\pi^0)$ must be within the range $0.7 < M(3\pi) < 2.0$ GeV/c^2.
- **LO events**: Events with no NLO photon, or where the NLO photon energy is below 0.05 GeV.
- **NLO LA events**: Events with a NLO photon of energy ≥ 0.05 GeV and $|\cos\theta_{\gamma_{\mathrm{NLO}}}| < 0.99$.
- **NLO SA events**: Events with a NLO photon of energy ≥ 0.05 GeV and $|\cos\theta_{\gamma_{\mathrm{NLO}}}| > 0.99$.

For each category, we determine the overall selection efficiency ε_i from the PHOKHARA MC sample, defined as the fraction of generator-level events in that category that pass the corresponding reconstruction and selection criteria described in the previous sections. These efficiencies are assumed to be consistent between data and MC.

The yields in data for each category ($N_i^{\text{Data, reco}}$) are obtained from the fits described in Sections 6.4.1 and 6.4.2 for NLO LA and SA categories, while the LO yield in data is taken from the nominal signal selection (4C fit $\chi^2 < 50$ with a single ISR photon). Using the MC-determined efficiencies, we can obtain the generator-level yields in data:

$$N_i^{\text{Data, gen}} = \frac{N_i^{\text{Data, reco}}}{\varepsilon_i}$$

where i denotes the event category (LO, NLO LA, or NLO SA). The total number of generated events in data is then the sum over all categories:

$$N_{\text{total}}^{\text{Data, gen}} = \sum_i N_i^{\text{Data, gen}}$$

The fraction of events in each category at generator level is then calculated as:

$$f_i^{\text{Data}} = \frac{N_i^{\text{Data, gen}}}{N_{\text{total}}^{\text{Data, gen}}}$$

These fractions represent the true production rates of LO, NLO LA, and NLO SA events in data, corrected for selection efficiencies.

For comparison, we directly compute the generator-level fractions from the PHOKHARA MC sample using the truth-level classification (including the same $M(3\pi)$ requirement):

$$f_i^{\text{MC}} = \frac{N_i^{\text{MC, gen}}}{N_{\text{total}}^{\text{MC, gen}}}$$

where $N_i^{\text{MC, gen}}$ are the counts of MC events in each truth category satisfying the $M(3\pi)$ requirement, and $N_{\text{total}}^{\text{MC, gen}}$ is the total number of generated MC events within the same $M(3\pi)$ range.

Table 6 presents the comparison between data and MC for the three categories. The statistical uncertainties on f_i^{Data} are propagated from the uncertainties on the fitted yields and the statistical uncertainties on the efficiencies. The uncertainties on f_i^{MC} are purely statistical from the MC sample size.

Table 6: Selection efficiencies and generator-level event fractions for LO, NLO LA, and NLO SA categories. The efficiencies are determined from PHOKHARA MC and represent the fraction of generator-level events that pass the corresponding reconstruction and selection criteria. The MC fractions are normalized such that $f_{\mathrm{LO}} + f_{\mathrm{NLO\,LA}} + f_{\mathrm{NLO\,SA}} = 100\%$.

Category	Efficiency (%)	f_i^{MC} (%)	f_i^{Data} (%)	c_{NLO}
LO	3.092 ± 0.006	76.879 ± 0.032	87.422 ± 0.903	1.137 ± 0.012
NLO LA	0.275 ± 0.006	6.543 ± 0.007	5.370 ± 0.852	0.821 ± 0.130
NLO SA	0.417 ± 0.004	16.578 ± 0.012	7.207 ± 0.471	0.435 ± 0.028

The ratios $f_i^{\mathrm{Data}}/f_i^{\mathrm{MC}}$ reveal a discrepancy between data and the PHOKHARA generator simulation. We observe that PHOKHARA overestimates the production rate of events with additional ISR photons, consistent with the findings from the BaBar study [15]. This discrepancy will be corrected in the unfolding procedure through the response matrix.

6.5 Correcting the signal yields

To construct the migration matrix for unfolding, we start from the signal MC samples that have been corrected for helix parameters and satisfy the $\chi^2_{4C} < 50$ selection requirement. The migration matrix element A_{ij} is intended to represent the number of events that truly belong to bin j and are reconstructed in bin i. The MC simulation does not perfectly describe data; discrepancies arise from the NLO ISR photon production rate and from the efficiencies of tracking, PID, π^0 reconstruction, and ISR photon detection. These must be corrected before the matrix can be used for unfolding.

Two types of corrections are applied to the MC events. The first correction addresses the discrepancy in the NLO ISR photon production rate. For each event from truth bin j, we assign a weight:

$$w_{\mathrm{NLO}} = c_{\mathrm{NLO}}(j),$$

where $c_{\mathrm{NLO}}(j)$ is the bin-dependent NLO correction factor.

The second correction accounts for the differences in reconstruction efficiencies. For each successfully reconstructed event, we assign an additional weight:

$$w_{\mathrm{eff}} = c_{\mathrm{trk}}(\pi^+) \cdot c_{\mathrm{pid}}(\pi^+) \cdot c_{\mathrm{trk}}(\pi^-) \cdot c_{\mathrm{pid}}(\pi^-) \cdot c_{\pi^0} \cdot c_{\gamma_{\mathrm{ISR}}}.$$

The corrected migration matrix is filled using the following prescription. Initially, all generated events are considered to be lost (i.e., placed into the missing category) with weight w_{NLO}. Then, for each reconstructed event that truly belongs to truth bin j and is observed in reconstructed bin i, we:

- Add $w_{\mathrm{NLO}} \cdot w_{\mathrm{eff}}$ to the migration matrix element A_{ij}^{corr};

- Subtract $w_{\mathrm{NLO}} \cdot w_{\mathrm{eff}}$ from the missing weight to remove the event's contribution from the missing category.

This procedure ensures that the total corrected weight $N^{\mathrm{corr}}_{\mathrm{total},j} = \sum_i A^{\mathrm{corr}}_{ij} + N^{\mathrm{corr}}_{\mathrm{miss},j}$ is conserved. The normalized response matrix P^{corr}_{ij} is then obtained by applying Eq. (8) to A^{corr}_{ij}.

7 Unfolding Procedure

7.1 The IDS unfolding method

To extract the generator-level yield $\left(\frac{dN}{dm}\right)_j^{\text{gen}}$ from the observed yields $\left(\frac{dN}{dm}\right)_i^{\text{obs}}$ and the response matrix P_{ij}, we employ the Iterative Dynamically Stabilized (IDS) unfolding method [13]. This section introduces the mathematical formulation of the method; the specific implementation, parameter optimization, and iteration strategy are described in the following sections.

From the MC simulation, we have constructed the migration matrix A_{ij} (shown in Fig. 12) and the response matrix P_{ij} normalized column-wise as in Eq. (8). The corresponding unfolding matrix is defined row-wise:

$$\tilde{P}_{ij} = \frac{A_{ij}}{\sum_{k=1}^{n_u} A_{ik}}, \tag{13}$$

which gives the probability that an event reconstructed in bin i originated from true bin j.

The IDS method uses a regularization function $f(|\Delta|, \tilde{\sigma}, \lambda)$ that dynamically distinguishes statistical fluctuations from genuine signals. Here, $|\Delta|$ represents the absolute difference between data and the model prediction (e.g., $|\Delta d_i|$ for the discrepancy in reconstructed bin i, or $|\Delta u_j|$ for the discrepancy in true bin j), $\tilde{\sigma}$ denotes the corresponding statistical uncertainty ($\tilde{\sigma} d_i$ or $\tilde{\sigma} u_j$), and λ is a tunable regularization parameter. The function satisfies the following asymptotic behavior:

$$\lim_{\frac{|\Delta|}{\lambda\tilde{\sigma}} \to 0} f(|\Delta|, \tilde{\sigma}, \lambda) = 0, \qquad \lim_{\frac{|\Delta|}{\lambda\tilde{\sigma}} \to \infty} f(|\Delta|, \tilde{\sigma}, \lambda) = 1. \tag{14}$$

Thus, for discrepancies much smaller than λ times the uncertainty, the function approaches zero, treating the difference as a fluctuation; for discrepancies much larger, it approaches unity, indicating a real signal that should be unfolded. The transition between these regimes is smooth, avoiding sharp cutoffs that could introduce artifacts.

In the IDS method, the unfolding is performed iteratively. At iteration k, the unfolded spectrum $u_j^{(k)}$ is given by:

$$u_j^{(k)} = \eta \cdot t_j + \sum_{i=1}^{n_d} \left[f\left(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U\right) \Delta d_i^{(k-1)} \tilde{P}_{ij}^{(k-1)} + \left(1 - f\left(|\Delta d_i^{(k-1)}|, \tilde{\sigma} d_i, \lambda_U\right)\right) \Delta d_i^{(k-1)} \delta_{ij} \right], \tag{15}$$

where:

- $u_j^{(k)}$ is the unfolded generator-level yield in bin j at iteration k,
- t_j is the true MC yield in bin j from simulation,
- $\eta = N_d^{\text{MC}} / N_{\text{MC}}$ is the normalization factor that scales the MC to match the data in regions without significant new structures,
- $\Delta d_i^{(k-1)} = d_i - B_i^d - \eta \cdot r_i^{(k-1)}$ is the difference between the observed data yield d_i and the sum of the estimated background B_i^d (which may be zero if no background subtraction is applied) and the normalized reconstructed MC spectrum $\eta \cdot r_i^{(k-1)}$,

- $r_i^{(k-1)}$ is the reconstructed MC spectrum at iteration $k-1$, obtained by folding the unfolded spectrum $u_j^{(k-1)}$ with the response matrix $P_{ij}^{(k-1)}$,
- $\tilde{\sigma} d_i$ is the statistical uncertainty of d_i,
- $\tilde{P}_{ij}^{(k-1)}$ is the unfolding matrix from the previous iteration,
- λ_U is the regularization parameter used in the unfolding step,
- δ_{ij} is the Kronecker delta, which equals 1 if $i = j$ and 0 otherwise.

The first term on the right-hand side, $\eta \cdot t_j$, represents the normalized true MC spectrum, which serves as a baseline. The remaining sum dynamically partitions the discrepancy $\Delta d_i^{(k-1)}$: a fraction f is unfolded using $\tilde{P}_{ij}^{(k-1)}$, while the complementary fraction $1-f$ is retained in the original bin $j = i$ via the Kronecker delta.

The response matrix itself can be iteratively improved. Using the difference $\Delta u_j = u_j - \eta \cdot t_j$ between the current unfolded result and the normalized true MC, the migration matrix is modified as:

$$A'_{ij} = A_{ij} + f\left(|\Delta u_j|, \tilde{\sigma} u_j, \lambda_M\right) \Delta u_j P_{ij} \frac{N_{\mathrm{MC}}}{N_d^{\mathrm{MC}}}, \quad \text{for } i \in \{1, \ldots, n_d\}, \tag{16}$$

where:

- Δu_j is the difference between the current unfolded yield u_j and the normalized true MC $\eta \cdot t_j$,
- $\tilde{\sigma} u_j$ is the statistical uncertainty associated with u_j,
- λ_M is the regularization parameter controlling the modification strength,
- P_{ij} is the current response matrix (column-normalized A_{ij}),
- $N_{\mathrm{MC}} = \sum_j t_j$ is the total number of generated MC events,
- N_d^{MC} is the number of data events compatible with the MC model (used for normalization).

The updated migration matrix A'_{ij} is then renormalized column-wise to obtain an improved response matrix P_{ij}, and row-wise to obtain an improved unfolding matrix $\tilde{P}_{ij}$.

7.2 Parameter optimization and Unfolding

The IDS method involves three regularization parameters: λ_L for the initial unfolding, λ_M for modifying the response matrix, and λ_U for subsequent unfoldings. The iterative procedure consists of the following three steps:

1. **Initial unfolding:** Perform an unfolding using Eq. (15) with $\lambda = \lambda_L$ and the initial response matrix P_{ij}, obtaining $u_j^{(0)}$.
2. **Matrix improvement:** Modify the migration matrix via Eq. (16) using λ_M and the current u_j. Renormalize to obtain updated P_{ij} and $\tilde{P}_{ij}$.

3. **Improved unfolding:** Perform an unfolding using Eq. (15) with $\lambda = \lambda_U$ and the updated matrices.

These three steps can be repeated for a chosen number of iterations.

In this analysis, the four parameters λ_M, λ_L, λ_U, and n_{iter} are optimized by scanning over a grid of possible values. For each parameter combination $(\lambda_M, \lambda_L, \lambda_U, n_{\text{iter}})$, we perform the IDS unfolding and evaluate the quality of the result using a χ^2 figure of merit. The χ^2 is computed by comparing the unfolded spectrum u_j to the true generator-level distribution t_j^{truth} from the MC simulation. The MC sample used for this comparison has been corrected following the procedure described in Sec. 6.5 and is normalized such that its total integral matches the total number of signal events observed in data (shown in Fig. 11), ensuring that the statistical uncertainties from the toy ensembles properly reflect the data statistics.

$$\chi^2 = \left(\mathbf{u} - \mathbf{t}^{\text{truth}}\right)^T \Sigma^{-1} \left(\mathbf{u} - \mathbf{t}^{\text{truth}}\right), \tag{17}$$

where $\mathbf{u}$ and $\mathbf{t}^{\text{truth}}$ are vectors of bin contents, and Σ is the covariance matrix of the unfolded spectrum. The covariance matrix accounts for the statistical uncertainties and their correlations introduced by the unfolding procedure. It is estimated using a MC sampling method:

1. Generate an ensemble of N_{toy} pseudo-data distributions by fluctuating the original observed spectrum according to its statistical uncertainties, using a random number generator.

2. For each pseudo-data distribution, perform the IDS unfolding with the same parameter combination $(\lambda_L, \lambda_U, n_{\text{iter}})$, obtaining an unfolded spectrum $u_j^{(k)}$ for toy k.

3. From the ensemble of unfolded spectra, compute the covariance matrix:

$$\Sigma_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} \left(u_i^{(k)} - \bar{u}_i\right)\left(u_j^{(k)} - \bar{u}_j\right), \tag{18}$$

where $\bar{u}_i = \frac{1}{N_{\text{toy}}} \sum_k u_i^{(k)}$ is the mean unfolded yield in bin i across the toy ensemble.

In cases where the covariance matrix is singular or ill-conditioned, a small diagonal perturbation $\lambda_{\text{reg}} I$ is added before inversion to ensure numerical stability, with $\lambda_{\text{reg}} = 10^{-6}$ in this analysis.

The optimal parameter combination is chosen based on several considerations: the number of iterations n_{iter} should be as small as possible to avoid unnecessary complexity; the χ^2 value defined in Eq. (17) should be small, indicating good agreement between the unfolded and true spectra; and the χ^2 per degree of freedom, χ^2/n_{dof}, should be close to or less than one, indicating that the uncertainties are well estimated.

It is emphasized that both the χ^2 and the uncertainty estimates are computed strictly within the three intervals and corresponding bin widths defined in Section 5.2. Since reconstructed events near the boundaries may have truth-level counterparts originating from outside the region of interest, we extend the boundaries for each mass region during the training of the unfolding procedure. This extension is applied solely to preserve the completeness of information within the defined region and does not affect the χ^2 or uncertainty calculations, which remain confined to the intervals in Section 5.2. Events falling outside the region of interest are not considered further.

Table 7: The resulting unfolding parameters for the three representative mass regions.

Region	λ_L	λ_U	λ_M	n_{iter}	χ^2/ndf			
					Round 03/04	Round 15	Round 16	Round 17
$\omega(782)$	0.1	0.1	1.0	3	67.75/40	20.74/40	23.81/40	11.97/40
$\phi(1020)$	0.1	0.1	1.0	3	1.09/30	11.46/30	10.70/30	9.52/30
ω'/ω''	0.1	0.1	1.0	3	0.90/76	2.71/76	10.11/76	1.11/76

Using the MC sample from Round 16 as a stand-in for data, a scan is performed over a grid of λ_L, λ_U, and n_{iter} values. The combination that best satisfies the above criteria is selected for the final unfolding of the data. The optimized parameters obtained from this procedure for the three representative mass regions are summarized in Table 7, along with the resulting χ^2/ndf values for all four data-taking rounds to demonstrate consistency.

To validate the stability of the unfolding procedure across different data-taking conditions, the same set of optimized parameters (from Table 7) is applied to the MC samples of other rounds: Round 03/04, Round 15, and Round 17. Figure 20 shows the unfolding results for each round and each mass region.

To further validate the unfolding procedure, we compare the relative statistical uncertainty of the unfolded spectrum with that of the test MC sample. The relative uncertainty of the test MC spectrum, δ_{test}, is computed by summing in quadrature the Poisson errors of each bin and dividing by the total yield:

$$\delta_{\mathrm{test}} = \frac{\sqrt{\sum_i \sigma_i^2}}{\sum_i N_i}, \tag{19}$$

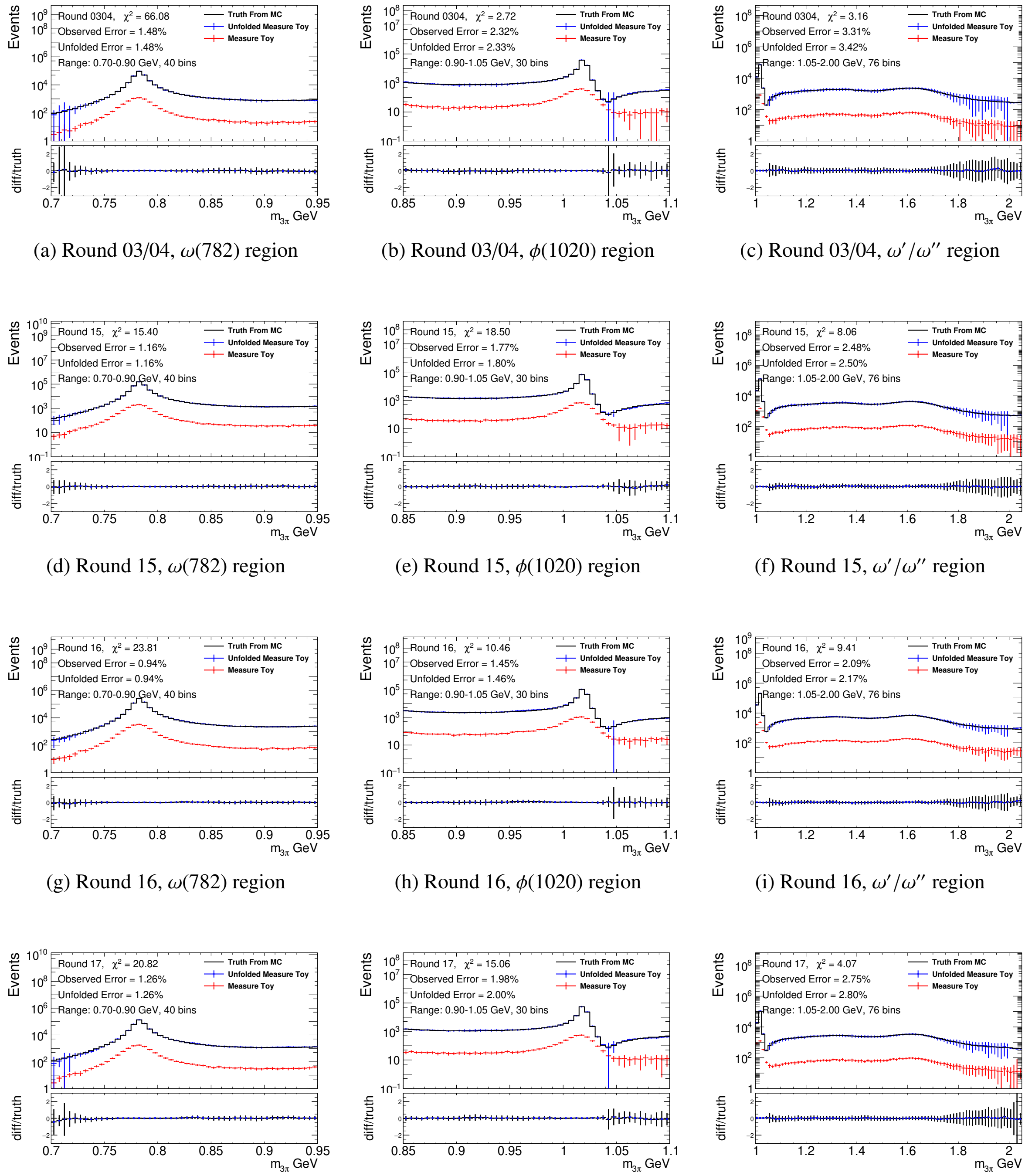
where N_i and σ_i are the content and Poisson error of bin i in the test MC distribution, respectively. This represents the Poisson statistical uncertainty of the test MC sample.

The relative uncertainty of the unfolded spectrum, $\delta_{\mathrm{unfolded}}$, is derived from the full covariance matrix Σ obtained from the toy MC ensemble:

$$\delta_{\mathrm{unfolded}} = \frac{\sqrt{\sum_{i,j} \Sigma_{ij}}}{\sum_i u_i}, \tag{20}$$

where u_i are the bin contents of the unfolded spectrum. The relative uncertainty of the unfolded spectrum should not be smaller than the Poisson uncertainty of the test MC sample.

Using the corrected migration matrix A_{ij}^{corr}, the unfolding of the $M_{3\pi}$ distribution is performed on data from Sec. 5.2. The unfolding procedure automatically accounts for the reconstruction and selection efficiency of the signal by incorporating information from missing events, rather than relying on the response matrix alone. The mass range for each resonance region is extended during the unfolding procedure and subsequently truncated back to the nominal interval defined in Tab. 7 for the final results. Figure 21 presents the unfolded $M_{3\pi}$ distributions for data in the three regions.



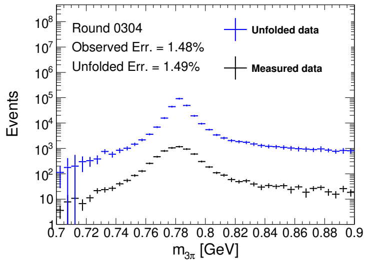
(a) Round 03/04, $\omega(782)$ region (b) Round 03/04, $\phi(1020)$ region (c) Round 03/04, ω'/ω'' region

(d) Round 15, $\omega(782)$ region (e) Round 15, $\phi(1020)$ region (f) Round 15, ω'/ω'' region

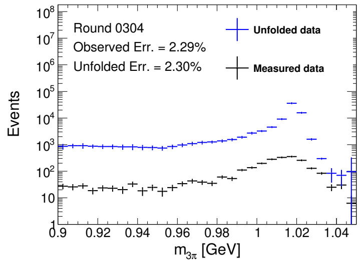
(g) Round 16, $\omega(782)$ region (h) Round 16, $\phi(1020)$ region (i) Round 16, ω'/ω'' region

(j) Round 17, $\omega(782)$ region (k) Round 17, $\phi(1020)$ region (l) Round 17, ω'/ω'' region

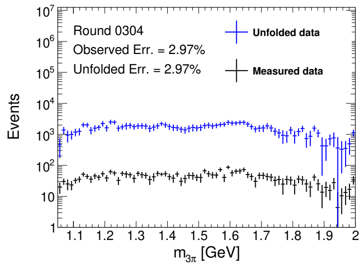
Figure 20: Unfolded results for four data-taking periods (Round 03/04, Round 15, Round 16, Round 17) using the IDS method. The unfolding parameters were optimized using the Round 16 MC sample (third row) and then applied to all rounds.



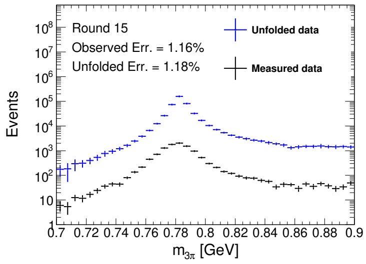
(a) Round 03/04, $\omega(782)$ region



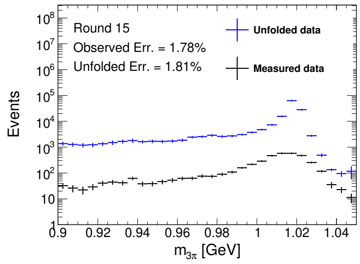
(b) Round 03/04, $\phi(1020)$ region



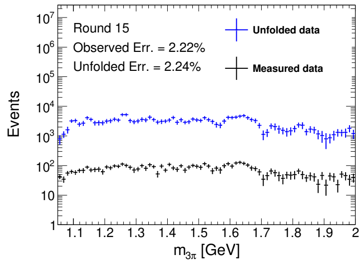
(c) Round 03/04, ω'/ω'' region



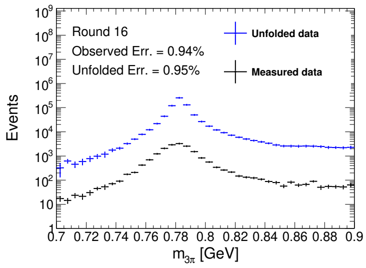
(d) Round 15, $\omega(782)$ region



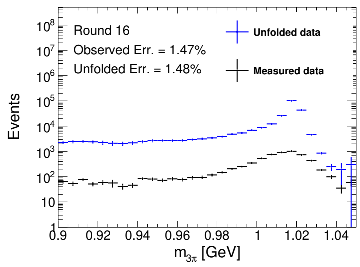
(e) Round 15, $\phi(1020)$ region



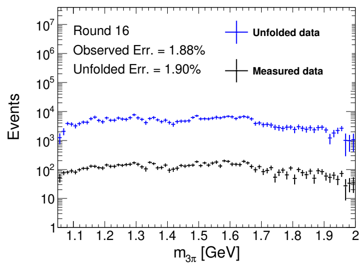
(f) Round 15, ω'/ω'' region



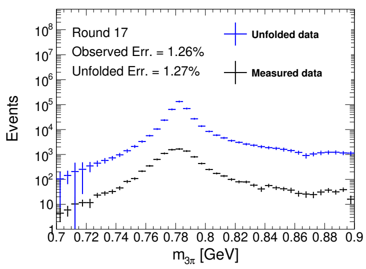
(g) Round 16, $\omega(782)$ region



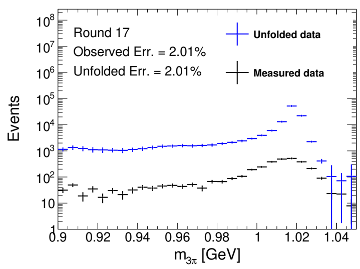
(h) Round 16, $\phi(1020)$ region



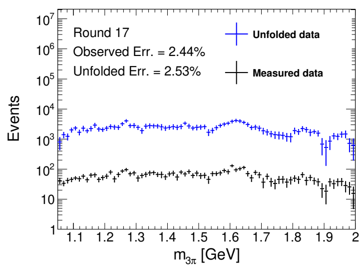
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 21: Unfolded $M_{3\pi}$ distributions after applying the corrected response matrix, shown for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). The results are shown only within the nominal mass ranges defined in Tab. 7.

8 Systematic Uncertainties

In this chapter, we discuss the systematic uncertainties considered in this analysis. The primary systematic uncertainty comes from the efficiency correction applied in the unfolding procedure. Additional uncertainties are associated with the requirement on E/p, the 4C kinematic fit, background estimate and the π^0 veto.

8.1 Efficiency correction uncertainties

The corrections applied to the MC events in the unfolding procedure are designed to mitigate discrepancies between data and simulation, including the NLO ISR photon production rate and the reconstruction efficiencies for tracking, PID, π^0 reconstruction, and ISR photon detection. The finite precision of these corrections introduces systematic uncertainties in the unfolded results.

The correction factors are grouped into seven independent sources:

$$c_{\text{NLO}}(j),\ c_{\text{trk}}(\pi^+),\ c_{\text{pid}}(\pi^+),\ c_{\text{trk}}(\pi^-),\ c_{\text{pid}}(\pi^-),\ c_{\pi^0},\ c_{\gamma_{\text{ISR}}}.$$

Each factor has an associated uncertainty, derived from auxiliary measurements or theoretical calculations. To propagate these uncertainties to the unfolded spectrum, we perform a toy MC procedure.

For each toy variation $k = 1, \ldots, N_{\text{toy}}$, we independently sample all seven correction factors according to their respective uncertainty distributions (Gaussian with widths given by the factor's uncertainty). This yields a set of seven parameters:

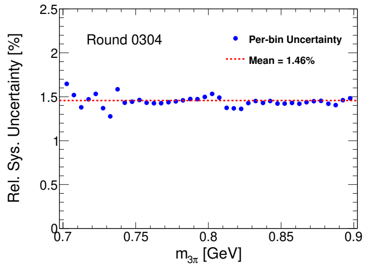
$$(c^{(k)}_{\text{NLO}}(j),\ c^{(k)}_{\text{trk}}(\pi^+),\ c^{(k)}_{\text{pid}}(\pi^+),\ c^{(k)}_{\text{trk}}(\pi^-),\ c^{(k)}_{\text{pid}}(\pi^-),\ c^{(k)}_{\pi^0},\ c^{(k)}_{\gamma_{\text{ISR}}}).$$

For each such parameter set, we recompute the corrected migration matrix $A^{\text{corr},(k)}_{ij}$ following the prescription in Sec. 6.5, then derive the normalized response matrix and perform the unfolding to obtain a toy unfolded spectrum $\{u^{(k)}_i\}$, where i indexes the truth bins.

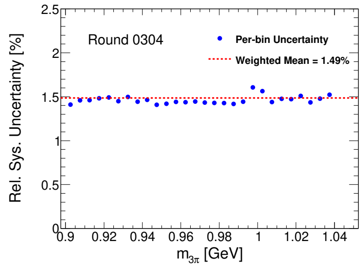
After processing all N_{toy} toys, the covariance matrix of the systematic uncertainty arising from the efficiency corrections is estimated as:

$$\Sigma^{\text{sys}}_{ij} = \frac{1}{N_{\text{toy}} - 1} \sum_{k=1}^{N_{\text{toy}}} \left(u^{(k)}_i - u^{\text{nom}}_i\right)\left(u^{(k)}_j - u^{\text{nom}}_j\right),$$

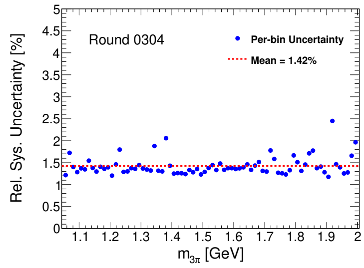
where u^{nom}_i is the nominal unfolded spectrum. The diagonal elements $\sqrt{\Sigma^{\text{sys}}_{ii}}$ provide the uncertainty per bin, while the off-diagonal elements quantify correlations between bins induced by the uncertainties in the correction factors. This covariance matrix is taken as the systematic uncertainty contribution from the efficiency corrections in the final results. This procedure is performed independently for each data-taking period (rounds 0304, 15, 16, and 17) and for each $M_{3\pi}$ region. The resulting diagonal systematic uncertainties per bin, expressed as relative uncertainties, are shown in Fig. 22 for all periods and regions.



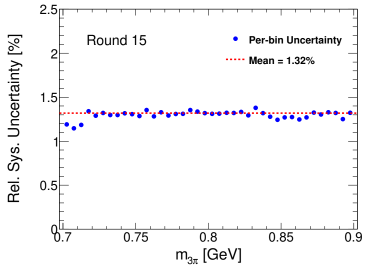
(a) Round 03/04, $\omega(782)$ region



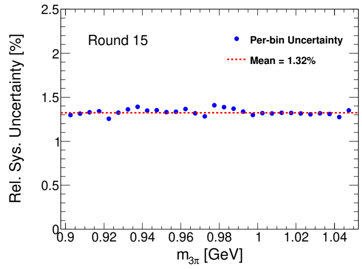
(b) Round 03/04, $\phi(1020)$ region



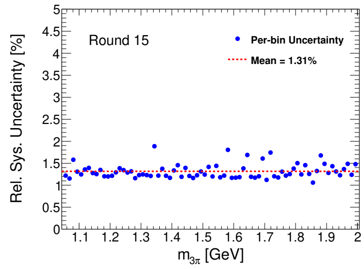
(c) Round 03/04, ω'/ω'' region



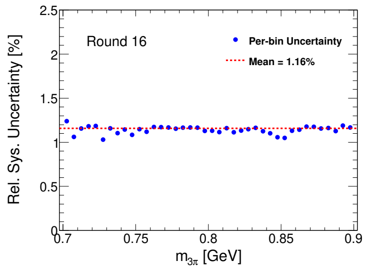
(d) Round 15, $\omega(782)$ region



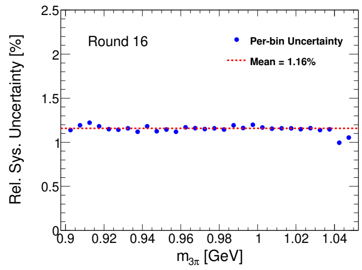
(e) Round 15, $\phi(1020)$ region



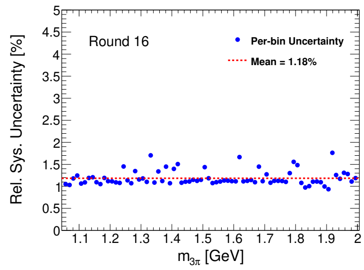
(f) Round 15, ω'/ω'' region



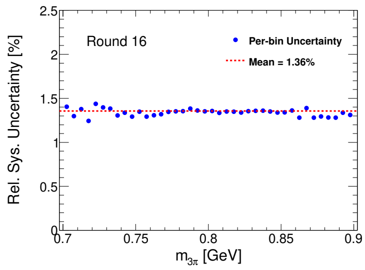
(g) Round 16, $\omega(782)$ region



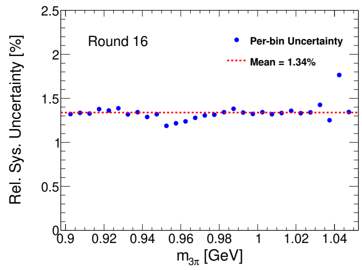
(h) Round 16, $\phi(1020)$ region



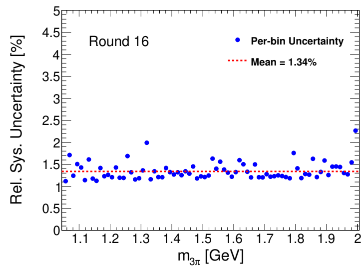
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 22: Diagonal relative systematic uncertainties (per bin) from efficiency corrections, shown for four data-taking periods. Each row corresponds to a different round: (a)–(c) Round 03/04, (d)–(f) Round 15, (g)–(i) Round 16, and (j)–(l) Round 17. The three columns represent different 3π invariant mass regions: $\omega(782)$ region (left), $\phi(1020)$ region (center), and higher-mass ω'/ω'' region (right). For each bin, the relative uncertainty is computed as $\sqrt{\Sigma_{ii}^{\text{sys}}}/u_i$, where u_i is the unfolded yield.

8.2 The detector resolution effects in unfolding

The migration of events between neighboring mass bins due to detector resolution is quantified by the response matrix P_{ij} constructed from MC simulations. As an example, Fig. 23 shows the distribution of reconstructed $m_{3\pi}$ for events generated in the truth bin $[0.785, 0.790]$ GeV$/c^2$ using the round 16 MC sample. A clear broadening is observed around the ω meson peak, which reflects the detector resolution as modeled in the MC simulation.

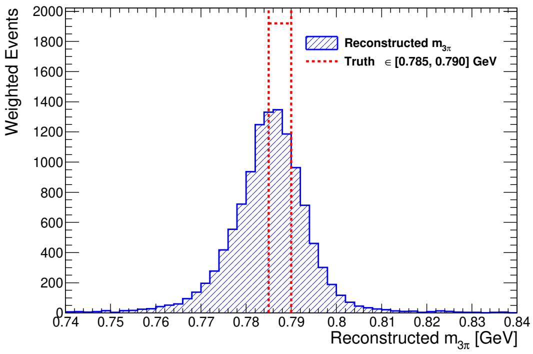


Figure 23: Distribution of reconstructed $m_{3\pi}$ for MC events generated in the truth bin $[0.785, 0.790]$ GeV$/c^2$ (round 16). The spread illustrates the detector resolution effect as modeled in MC, which causes migrations to neighboring bins.

The observed yield in each mass bin is related to the generator-level yield through the response matrix:

$$\left(\frac{dN}{dm}\right)_i^{\text{obs}} = \sum_j P_{ij} \left(\frac{dN}{dm}\right)_j^{\text{gen}} \tag{21}$$

where P_{ij} is constructed entirely from MC simulations. Both data and MC events suffer from detector resolution broadening, but the response matrix only contains the MC-predicted migration pattern. A key concern is whether the detector resolution in data differs from that in MC, as any mismatch would bias the unfolded cross section.

To quantify this potential discrepancy, we introduce a calibrated response matrix $P_{ij}^{\text{cal}}(\delta, \kappa)$ that parameterizes the differences between data and MC. For each truth bin j, the mean and standard deviation of the reconstructed mass distribution from MC are defined as:

$$\mu_{0,j} = \frac{\sum_i m_i P_{ij}}{\sum_i P_{ij}}, \qquad \sigma_{0,j} = \sqrt{\frac{\sum_i (m_i - \mu_{0,j})^2 P_{ij}}{\sum_i P_{ij}}} \tag{22}$$

where m_i is the center of reconstructed bin i. The data response is then modeled by applying a global shift δ and a broadening factor κ to these MC-derived parameters:

$$\mu_j = \mu_{0,j} + \delta, \qquad \sigma_j = \kappa \sigma_{0,j} \tag{23}$$

These calibrated parameters define a Gaussian smearing kernel $G_{ik}^{(j)}(\delta, \kappa)$ for each truth bin j, which is then convolved with the original MC response matrix to obtain the calibrated transfer matrix:

$$P_{ij}^{\mathrm{cal}}(\delta,\kappa) = \sum_k G_{ik}^{(j)}(\delta,\kappa) P_{kj}^{(0)} \tag{24}$$

where $P_{kj}^{(0)}$ is the nominal MC response matrix. The observed data distribution is then modeled as:

$$\left(\frac{dN}{dm}\right)_i^{\mathrm{obs}} = \sum_j P_{ij}^{\mathrm{cal}}(\delta,\kappa) \left(\frac{dN}{dm}\right)_j^{\mathrm{VMD}} \tag{25}$$

By fitting Eq. (25) to the observed data using the known VMD lineshape as the true spectrum, the fitted parameters δ and κ directly quantify the shift and broadening differences between data and MC in the response matrix. This approach provides a data-driven estimate of the systematic uncertainty arising from detector resolution effects.

8.3 The requirement on E/p

To evaluate the systematic uncertainty associated with the $E/p < 0.8$ requirement in the $e^+e^- \to \pi^+\pi^-\pi^0$ process, we adopt a linear approximation as a function of the 3π invariant mass $M_{3\pi}$. The motivation is that as $M_{3\pi}$ increases, the kinematic range of the individual π^+ or π^- momenta broadens significantly, which in turn affects the response of the E/p selector differently in data and MC. Therefore, the data–MC efficiency ratio of the E/p cut is expected to vary smoothly with $M_{3\pi}$.

We calibrate this dependence using two control samples. The first is the $\omega \to \pi^+\pi^-\pi^0$ region at low $M_{3\pi}$, where we use the selection criteria and samples described in Sec. 6.2 to select clean $e^+e^- \to \omega\pi^0$ events. The efficiency of the $E/p < 0.8$ requirement is defined as the fraction of events retained after applying the cut. The relative difference between data and MC is quantified as:

$$\delta_{E/p} = \frac{\varepsilon_{\mathrm{Data}}}{\varepsilon_{\mathrm{MC}}} - 1.$$

Table 8 summarizes the results for the Round 16 dataset in the ω signal region.

Table 8: The $E/p < 0.8$ efficiencies in data and MC for the ω signal region, and the relative difference $\varepsilon_{\mathrm{Data}}/\varepsilon_{\mathrm{MC}} - 1$ (Round 16).

Sample	$N_{\text{without } E/p}$	$N_{\text{with } E/p}$	ε
Data	20294.00	18165.00	0.8951
MC	19949.43	17878.60	0.8962
$\varepsilon_{\mathrm{Data}}/\varepsilon_{\mathrm{MC}} - 1 = -0.123\%$			

For the high-$M_{3\pi}$ region, we use the control sample of $e^+e^- \to \pi^+\pi^-\pi^0$ near the J/ψ mass, following the analysis in Ref. [18]. From that study, the relative difference between data and MC due to the E/p requirement is:

$$\delta_{E/p}^{J/\psi} = \frac{0.9}{93.28} = +0.965\%.$$

Assuming a linear dependence of $\delta_{E/p}$ on $M_{3\pi}$ between the ω and J/ψ reference points, the systematic uncertainty for an intermediate $M_{3\pi}$ is obtained by linear interpolation:

$$\delta_{E/p}(M_{3\pi}) = \delta_{E/p}^{\omega} + \frac{\delta_{E/p}^{J/\psi} - \delta_{E/p}^{\omega}}{M_{J/\psi} - M_{\omega}} \cdot (M_{3\pi} - M_{\omega}),$$

where $M_\omega = 0.782$ GeV and $M_{J/\psi} = 3.097$ GeV. For $M_{3\pi}$ below the ω or above the J/ψ, we conservatively take the value of the nearest reference point (constant extrapolation). The resulting data–MC difference in percentage is then taken as the systematic uncertainty for the given 3π mass bin.

This percentage directly applies to the generated spectrum $\left(\frac{dN}{dm}\right)^{\text{gen}}$, since the efficiency ε is already incorporated in the unfolding procedure, as first shown in Eq. 6:

$$\left(\frac{dN}{dm}\right)^{\text{gen}} = \sigma_{3\pi}^{\text{dress}}(m) \cdot \frac{dL}{dm} \cdot \varepsilon \cdot \mathcal{B}(\pi^0 \to \gamma\gamma).$$

Unlike the statistical uncertainty from unfolding, which yields a correlated covariance matrix, this systematic source is assumed to be uncorrelated among mass bins. Therefore, it is implemented as a diagonal covariance matrix and added in quadrature to the total covariance matrix of the cross section measurement.

8.4 The 4C kinematic fit

The nominal results for both the measured spectrum $\left(\frac{dN}{dm}\right)$ and the response matrix used in the unfolding procedure are obtained with the $\chi^2_{\text{corr}} < 50$ requirement applied in the 4C kinematic fit. To evaluate the associated systematic uncertainty, we repeat the entire analysis chain — including the estimation of $\left(\frac{dN}{dm}\right)$ and the unfolding procedure — after disabling the correction of the helix parameters for charged tracks in the kinematic fit.

In principle, one could vary the helix parameter corrections according to their uncertainties and repeat the unfolding multiple times to extract the covariance matrix, as described in Sec. 8.1. However, the uncertainties on these corrections are very small, and such an approach would likely lead to an underestimate of the systematic uncertainty. To be conservative, we instead take half of the difference between the unfolded spectra obtained with the alternative and nominal configurations as the systematic uncertainty for each bin. To account for bin-to-bin correlations, we construct the full covariance matrix as described below.

Let Δ_i be the difference between the alternative and nominal unfolded spectra in bin i:

$$\Delta_i = \left(\frac{dN}{dm}\right)_i^{\text{alt}} - \left(\frac{dN}{dm}\right)_i^{\text{nom}} \tag{26}$$

The diagonal elements of the covariance matrix are given by:

$$\text{Cov}_{ii} = \left(\frac{1}{2}\Delta_i\right)^2 \tag{27}$$

The off-diagonal elements are constructed as:

$$\mathrm{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\mathrm{Cov}_{ii} \cdot \mathrm{Cov}_{jj}},$$

where $s_i = \mathrm{sign}(\Delta_i)$.

This procedure yields a complete covariance matrix that accounts for bin-to-bin correlations: bins with the same sign are assumed to be fully positively correlated (+1), while bins with opposite signs are fully anti-correlated (−1). The resulting covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

8.5 Background estimate

The background in the $\pi^+\pi^-\pi^0$ sample consists mainly of two components. The dominant background arises from the $e^+e^- \to \pi^+\pi^-\pi^0\pi^0$ process. Its contribution is estimated using a data-driven method, where the background level in each 3π mass bin is evaluated and accompanied by a statistical uncertainty. In the fit to extract the 3π yield, this background component is constrained via a Gaussian penalty term that incorporates its uncertainty. Therefore, no additional systematic uncertainty is assigned for this dominant background.

For all other remaining background processes, a first- or second-order polynomial is used in the nominal fit, with all parameters left floating. The main systematic uncertainty associated with these backgrounds comes from their shape. To evaluate this effect, we perform an alternative fit in which the background shape is taken from the inclusive MC sample, and the normalization is constrained via a Gaussian term fixed to the expected yield based on the integrated luminosity. The resulting 3π yield after the alternative fit is then propagated through the full unfolding procedure.

Following the same procedure described in Sec. 8.4, we construct a covariance matrix using Δ_i (the difference between the alternative and nominal unfolded spectra) instead of $\frac{1}{2}\Delta_i$ for the diagonal elements. The diagonal elements of the covariance matrix are given by:

$$\mathrm{Cov}_{ii} = \Delta_i^2$$

The off-diagonal elements are constructed as:

$$\mathrm{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\mathrm{Cov}_{ii} \cdot \mathrm{Cov}_{jj}},$$

where $s_i = \mathrm{sign}(\Delta_i)$. This covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

8.6 The π^0 veto

The π^0 veto condition is applied only in the high-mass region of the spectrum. To evaluate the associated systematic uncertainty, we repeat the analysis — including the unfolding procedure — after removing this veto condition, focusing specifically on the third mass interval where the veto is originally applied.

Following the same procedure described in Sec. 8.4, we construct a covariance matrix using Δ_i (the difference between the unfolded spectra obtained without and with the π^0 veto) for the diagonal elements:

$$\mathrm{Cov}_{ii} = \Delta_i^2$$

The off-diagonal elements are constructed as:

$$\mathrm{Cov}_{ij} = s_i \cdot s_j \cdot \sqrt{\mathrm{Cov}_{ii} \cdot \mathrm{Cov}_{jj}},$$

where $s_i = \mathrm{sign}(\Delta_i)$. This covariance matrix is then added in quadrature to the total covariance matrix of the cross section measurement.

9 Summary

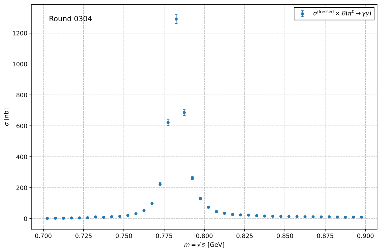
9.1 Calculation of cross sections

From the unfolding procedure described in the previous section, we obtain the unfolded $M_{3\pi}$ mass spectrum $\left(\frac{dN}{dm}\right)^{\text{gen}}$ for each data-taking period. The relationship between the unfolded mass spectrum and the dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ is given by:

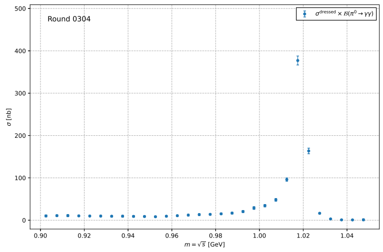
$$\left(\frac{dN}{dm}\right)^{\text{gen}} = \sigma_{3\pi}^{\text{dress}}(m) \cdot \frac{dL}{dm} \cdot \mathcal{B}(\pi^0 \to \gamma\gamma) \tag{28}$$

where $\frac{dL}{dm}$ is the effective luminosity per unit mass, and $\mathcal{B}(\pi^0 \to \gamma\gamma) = (98.823 \pm 0.034)\%$ is the branching fraction of $\pi^0 \to \gamma\gamma$ as quoted by the Particle Data Group (PDG) [17]. The complete formula for calculating $\frac{dL}{dm}$ is provided in Sec. 5.

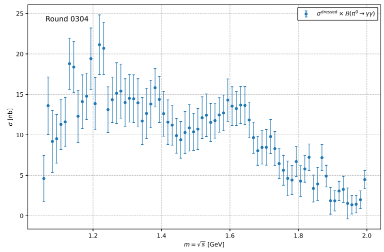
Using Eq. 28, we obtain four independent dressed cross section results $\sigma_{3\pi}^{\text{dress}}(m) \cdot \mathcal{B}(\pi^0 \to \gamma\gamma)$ corresponding to the four data-taking periods: Round 03/04, Round 15, Round 16, and Round 17. These results are presented in Fig. 24. For each period, the unfolding procedure yields not only the unfolded mass spectrum but also its corresponding covariance matrix, which fully describes the uncertainties and correlations of $\left(\frac{dN}{dm}\right)^{\text{gen}}$. Consequently, all subsequent uncertainties in the dressed cross section are propagated from these covariance matrices.



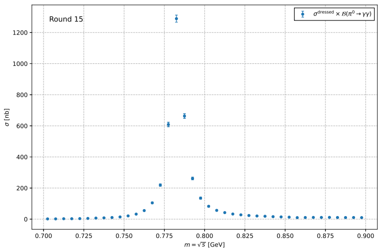
(a) Round 03/04, $\omega(782)$ region



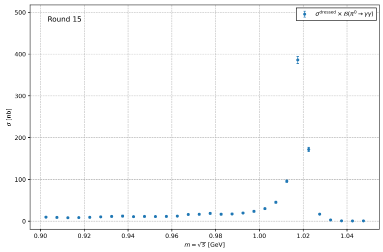
(b) Round 03/04, $\phi(1020)$ region



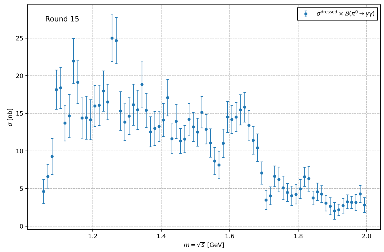
(c) Round 03/04, ω'/ω'' region



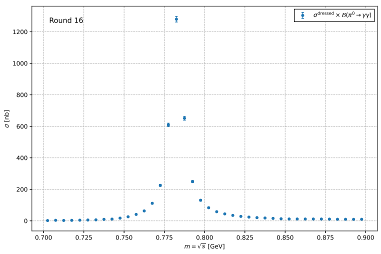
(d) Round 15, $\omega(782)$ region



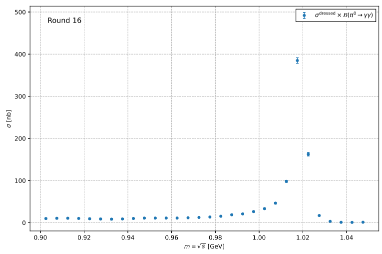
(e) Round 15, $\phi(1020)$ region



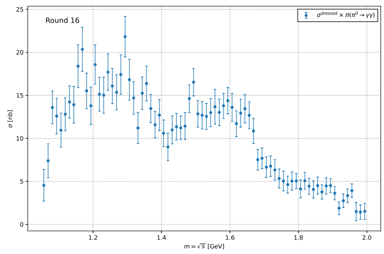
(f) Round 15, ω'/ω'' region



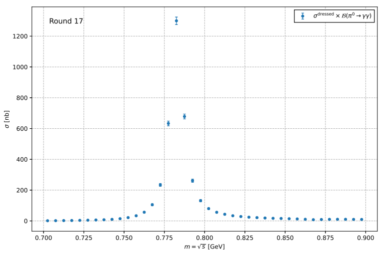
(g) Round 16, $\omega(782)$ region



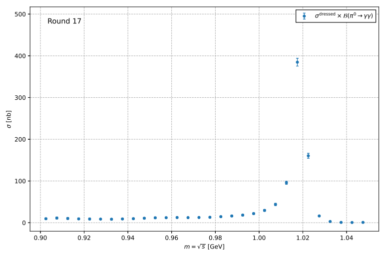
(h) Round 16, $\phi(1020)$ region



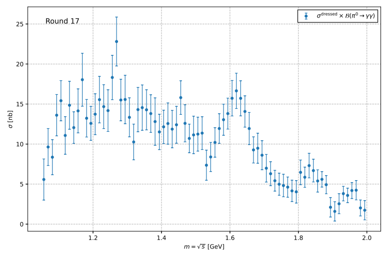
(i) Round 16, ω'/ω'' region



(j) Round 17, $\omega(782)$ region



(k) Round 17, $\phi(1020)$ region



(l) Round 17, ω'/ω'' region

Figure 24: Dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ results for the four independent data-taking periods (Round 03/04, Round 15, Round 16, and Round 17). The uncertainties are derived from the covariance matrices obtained from the unfolding procedure, which fully describe the correlations of the unfolded mass spectrum.

9.2 Combination of independent dressed cross section measurements

Since the four data-taking periods (Round 03/04, Round 15, Round 16, and Round 17) are statistically independent, we can combine their dressed cross section results to obtain a more precise measurement. However, a simple weighted average using only the diagonal uncertainties would be suboptimal, as it neglects the bin-to-bin correlations within each measurement that arise from the unfolding procedure.

To properly account for these correlations, we employ the Best Linear Unbiased Estimator (BLUE) method [**?**, **?**]. BLUE provides a systematic framework for combining multiple correlated measurements that estimate the same underlying quantity, producing an estimate that is both unbiased and has minimum variance.

9.2.1 Formalism of BLUE

Suppose we have n independent measurements $\mathbf{x}_i$ ($i = 1, \ldots, n$), each being a vector of length m (corresponding to the m invariant mass bins). Each measurement is an unbiased estimator of the true value $\boldsymbol{\mu}$:

$$E[\mathbf{x}_i] = \boldsymbol{\mu}, \quad \mathrm{Cov}(\mathbf{x}_i) = \mathbf{V}_i \tag{29}$$

where $\mathbf{V}_i$ is the $m \times m$ covariance matrix for the i-th measurement, which contains both the variances ($\sigma_{i,p}^2$ on the diagonal) and the bin-to-bin covariances ($\rho_{pq}\sigma_{i,p}\sigma_{i,q}$ off-diagonal). Since the measurements are independent, the cross-covariances satisfy $\mathrm{Cov}(\mathbf{x}_i, \mathbf{x}_j) = 0$ for $i \neq j$.

We seek a linear combination of all measurements:

$$\hat{\boldsymbol{\mu}} = \sum_{i=1}^{n} \mathbf{W}_i \mathbf{x}_i \tag{30}$$

where each $\mathbf{W}_i$ is an $m \times m$ weight matrix to be determined. The estimator $\hat{\boldsymbol{\mu}}$ is required to be unbiased:

$$E[\hat{\boldsymbol{\mu}}] = \sum_{i=1}^{n} \mathbf{W}_i E[\mathbf{x}_i] = \left(\sum_{i=1}^{n} \mathbf{W}_i\right) \boldsymbol{\mu} = \boldsymbol{\mu} \tag{31}$$

which imposes the constraint $\sum_{i=1}^{n} \mathbf{W}_i = \mathbf{I}$ (the identity matrix).

The covariance matrix of the combined estimator is:

$$\mathbf{U} = \mathrm{Cov}(\hat{\boldsymbol{\mu}}) = \sum_{i=1}^{n} \mathbf{W}_i \mathbf{V}_i \mathbf{W}_i^T \tag{32}$$

since the cross-terms vanish due to independence.

9.2.2 Optimal weights

The BLUE method minimizes the variance (in the sense of matrix trace or any positive-definite metric) subject to the unbiasedness constraint. Solving this optimization problem using Lagrange multipliers yields the optimal weight matrices:

$$\mathbf{W}_i = \left(\sum_{j=1}^{n} \mathbf{V}_j^{-1}\right)^{-1} \mathbf{V}_i^{-1} \tag{33}$$

provided that all $\mathbf{V}_i$ are invertible. If some covariance matrices are singular or near-singular, we use the Moore-Penrose pseudo-inverse instead.

Substituting these weights back gives the combined estimate:

$$\hat{\boldsymbol{\mu}} = \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1}\right)^{-1} \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1}\mathbf{x}_i\right) \tag{34}$$

and its covariance matrix:

$$\mathbf{U} = \left(\sum_{i=1}^{n} \mathbf{V}_i^{-1}\right)^{-1} \tag{35}$$

9.2.3 Application to this analysis

We apply the BLUE method to combine the four independent measurements of $\sigma_{3\pi}^{\text{dress}}(m) \cdot \mathcal{B}(\pi^0 \to \gamma\gamma)$, each represented by a vector $\mathbf{x}_i$ (size 40 for Region 1, 30 for Region 2, and 76 for Region 3) and its associated covariance matrix $\mathbf{V}_i$ obtained from the unfolding procedure via standard error propagation. The four $\mathbf{V}_i$ are derived from the covariance matrices of the unfolded mass spectra $\left(\frac{dN}{dm}\right)^{\text{gen}}$, scaled by the luminosity factors:

$$\mathbf{V}_i = \frac{\mathbf{V}_i^{(N)}}{\Delta L_i^2} \tag{36}$$

where $\mathbf{V}_i^{(N)}$ is the covariance matrix of the unfolded yield, and ΔL_i is the integrated luminosity per mass bin for the i-th period. Note that the branching fraction $\mathcal{B}(\pi^0 \to \gamma\gamma)$ is intentionally omitted here, as it will be applied after the combination to allow for separate treatment of its systematic uncertainty.

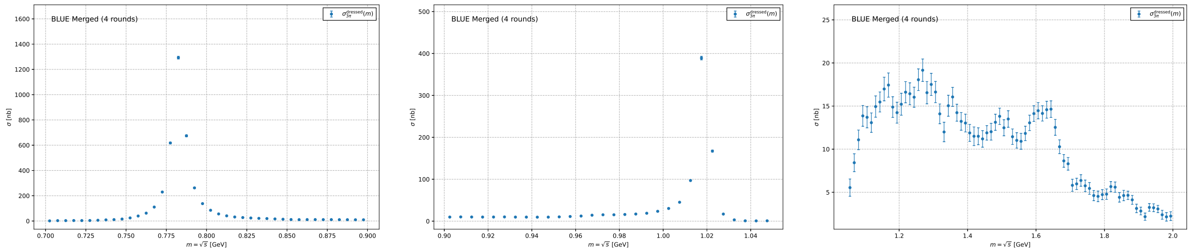
The combined result $\hat{\boldsymbol{\mu}}_{\text{withBR}}$ therefore represents the best estimate of $\sigma_{3\pi}^{\text{dress}}(m) \cdot \mathcal{B}(\pi^0 \to \gamma\gamma)$, with its covariance matrix $\mathbf{U}$ given by Eq. (35). The final dressed cross section is obtained by dividing by the central value of the branching fraction:

$$\hat{\boldsymbol{\mu}}_{\text{dressed}} = \frac{\hat{\boldsymbol{\mu}}_{\text{withBR}}}{\mathcal{B}(\pi^0 \to \gamma\gamma)}, \qquad \mathbf{U}_{\text{dressed}} = \frac{\mathbf{U}}{\mathcal{B}(\pi^0 \to \gamma\gamma)^2} \tag{37}$$

The covariance matrix $\mathbf{U}_{\text{dressed}}$ accounts for all statistical uncertainties propagated from the unfolding procedure. The systematic uncertainty due to the branching fraction is not included here and will be treated separately in Sec. **??**, as the final dressed cross section is not the ultimate physical quantity of interest in this analysis.

Figure 25 shows the BLUE-combined dressed cross section results for the three invariant mass regions: the $\omega(782)$ region (0.7 – 0.9 GeV), the $\phi(1020)$ region (0.9 – 1.05 GeV), and the ω'/ω'' region (1.05 – 2.0 GeV). The combined result exhibits significantly improved precision compared to individual measurements, particularly in the $\omega(782)$ region where the statistical uncertainties dominate.

For completeness, Fig. 26 presents the combined results across all three regions on a logarithmic



(a) $\omega(782)$ region (b) $\phi(1020)$ region (c) ω'/ω'' region

Figure 25: BLUE-combined dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ results for the three invariant mass regions. The error bars represent the statistical uncertainties propagated from the covariance matrices. The combination significantly improves the statistical precision compared to any single data-taking period.

scale, illustrating the full spectrum from 0.7 to 2.0 GeV in a single view.

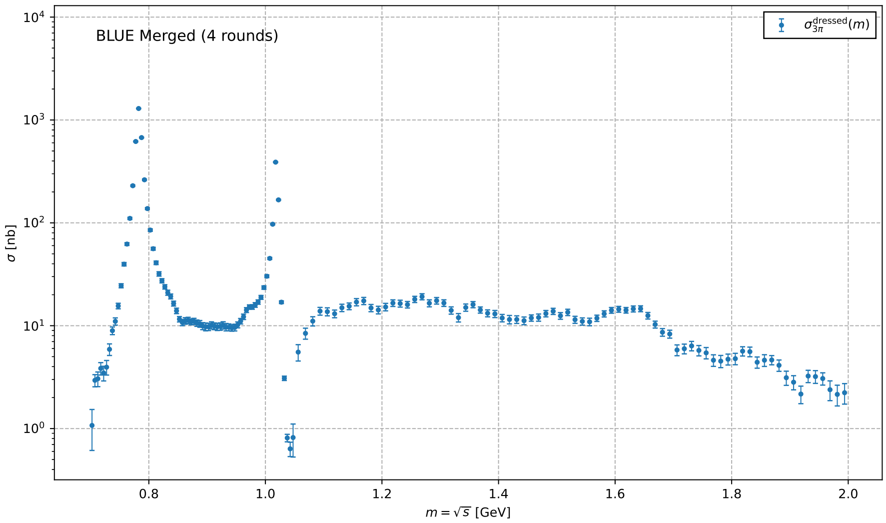


Figure 26: BLUE-combined dressed cross section $\sigma_{3\pi}^{\text{dress}}(m)$ across all three invariant mass regions, shown on a logarithmic scale.

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APPENDIX